

REPORT

SKINCARE ROUTINE RISK ASSESSMENT SYSTEM

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Chapter 1:

Project Overview

1.1 Context and Introduction

Skincare has become a daily habit for millions of people. Social media platforms such as Instagram and TikTok constantly promote new products, trends, and routines. While these trends make skincare exciting and accessible, they also create confusion and misuse

Many users , especially young ones , apply multiple products without understanding ingredient interactions, frequency limits, or whether the product matches their skin type. As a result, instead of achieving healthy skin, they experience irritation, acne, redness, or even long-term damage.

To better understand the problem, we introduced the story of Sarra.

Sarra follows beauty influencers online. Each time a new product becomes popular, she buys it. She layers serums, creams, exfoliants, and cleansers together without understanding:

- How the ingredients interact with each other
- How often they should be used
- Whether they match her skin type

After a few weeks, her skin starts reacting some acne appears , redness increases and dryness becomes noticeable

Sarra is just one of many users who misuse skincare products and put their skin at risk , those users :

- Mix products without knowing ingredient interactions
- Follow trends instead of their real skin needs
- Use active ingredients too often
- Have difficulty understanding usage instructions
- Try multiple new products frequently
- Use products that do not match their skin type

This shows that the problem is not the products themselves, but how they are used that's why our project is about an AI-based skincare routine evaluation system that helps users assess the safety of their skincare routine before damage occurs

1.2 Domain

This project belongs to the **health domain**, more specifically **dermatology and preventive skincare**.

However, our goal is not to replace dermatologists or provide medical diagnosis. Instead, our objective is to:

- Reduce unsafe product usage
- Prevent long-term skin damage
- Educate users
- Promote safe skincare habits

We focus on **prevention and awareness**.

1.3 Problematic

Instead of reacting after skin damage already appears, the key question of this project asks is :

Can we evaluate the skincare product ingredient interaction and suitability before it is used in order to prevent irritation and long-term damage?

From this main question, we identified three main problems:

1.3.1 Ingredient Interaction Risk

Many users mix products without understanding how their ingredients interact. In some cases, ingredients are safe when used alone but when we combine them they can cause skin damages

So we ask:

Can we predict incompatible ingredient combinations before irritation occurs?

1.3.2 Ingredient Compatibility with skin type

Even if individual products are safe, it's important that the products are suitable for the user's skin type .

For this reason, we ask:

Can we predict the skin type a skincare product is suitable for based on its ingredients?

1.3.3 Ingredient use for concerns

Product suitability for skin type is not the only important factor, but also the main skin concern that the product addresses.

The question is :

Can we predict the main skin concern targeted by a skincare product based on its ingredients?

1.4 Study of the existing

Before starting any project, it's important to make an analysis of what already exists in the market . In the skin products industry there is a lot of competitor , different applications were done to check product, give it a score ,check

ingredients ,check compatibility with skin type and many other applications each one have different focus ,we took a few examples and we examined them so they can assist us in developing a solution tailored to our needs.

In the table Below some Examples of applications their functionalities

Application	Definition	Limitation
Ceyra	a skincare scanner that helps you check if skincare products are good for your unique skin type Main functionalities : → smart routines → ingredient breakdown → skin profile → managements of personal product	→ Limited Diagnostic → Risk of Misleading Users → Privacy & Data Concerns → Few user reviews / uncertain reliability → Basic feature and lack of personalization
incibeauty	INCI Beauty helps users navigate the vast array of ingredients used in cosmetics Main functionalities : → Product scanning → Ingredient details → search for products → Product ratings → Alternative recommendations	→ Ads Issues → Rating can be misleading and biased → focus on products and not user → UI is confusing
SkinSAFE	SkinSAFE is a skincare product checking app that helps users figure out whether a product's ingredients are safe for their specific skin needs or sensitivities Main functionalities : → Personalized profiles → Product Search & Barcode Scanning → Get personalized product recommendations	→ accuracy problems → limited search result → Not very user friendly → Subscription are expensive
skinSort	A platform designed to help users analyze cosmetic ingredients, track routines, and find products tailored to their specific skin type and goals Main functionalities : → Ingredient Analysis and Scanner → Personalized Match Scores → Routine Tracker and Creator	→ app can be buggy failing to load pages → confusing UI → Ingredient interactions are not taken into consideration → Data errors → routine and recommendation suggesting clashing products

	→ Product Discovery and Comparison → Community and Insights	→ lack of safety of data
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Even though these apps offer different functionalities, Sometimes the apps are more biased toward reviews and product popularity. User interfaces are mostly confusing and hard to use. Ingredient interactions are mostly not taken into consideration, which can put users at risk. That is why our application solves these problems by providing a user-friendly interface that offers clear information and statistics while focusing on the product's ingredients rather than the product as a whole. This makes it more personalized, since it helps users know whether a product is suitable for their skin type, understand which skin concern the product is meant to address, and check if the ingredients within one product or between multiple products are compatible and safe to use together without causing danger or damaging the skin.

1.5 Business Hypotheses

To design our system, we based it on several hypotheses:

- Ingredient combinations affect safety
- Skin type influences risk
- Ingredients influence skin type suitability
- Ingredients are specified for specific concerns
- Products can be suitable for more than 1 skin type
- Two good products do not necessarily mean that their ingredients interact well together.

→ These hypotheses helped us to structure our AI model logic

1.6 Proposed Solution

We propose an **intelligent skincare routine system** that analyzes product ingredients to check the safety of use and suitability for the users skin and concern,

Our application collects two main types of data:

1. Dermatological Products

- Product names & brands
- Skin type
- Skin concern
- Product type

2. Product and Usage Behavior

- Product names and ingredients
- Frequency of using the products
- Duration of using each product and when it 's used (morning / evening)
- Ordering of application

How Our Application Works ?

The user will have two separate parts in the system one for ingredient compatibility check , the other for skin type and concern suitability

1. Ingredient Compatibility Detection :

First, the system checks whether the ingredients in the routine are safe to use together.

It identifies:

- Ingredient combinations that may cause irritation
- Ingredients that are not suitable for the user's skin type or sensitivity

This helps users avoid harmful combinations without needing to understand complex ingredient labels

2. Skin type suitability & concern addressed :

The users enters a picture or text of the ingredients of a certain skincare product ,the system give these results

- Skin type the product works with

- Skin concern the product address

Instead of showing complicated information, the app summarizes everything into simple results . This makes it easier for users to understand the safety of the ingredients and the products suitability.

1.7 Objectives

The main objectives of this project are:

- Reduce unsafe product usage among everyday skincare users
- Prevent long-term skin damage through early detection of risky combinations
- Educate users about ingredient interactions in a simple way
- Promote safe and healthy skincare habits
- Use of suitable products based on the user profile
- Using the right product for the right concern

Note: This project does not aim to replace dermatologists or provide medical diagnosis. It focuses on prevention and awareness.

1.8 Work Methodology: CRISP-DM

For our project we are going to use **CRISP-DM** which is a cyclical process that provides a structured approach to planning, organizing, and implementing a data mining project

Our Project is going to be divided into 6 phases which are

1. Business understanding
2. Data Understanding
3. Data Preparation
4. Modeling
5. Evaluation
6. Deployment

1.Business understanding

This phase focuses on understanding the objectives and requirements of the project

In this phase, we're going to start by identifying the objective of what we want to accomplish through this project, which is helping users choose their products and decrease risk. We will define the success criteria like accurate product analysis and recommendations, as well as assessing the current situation by analyzing what already exists in the market and what can be done to improve and add to it. This phase includes as well setting our goal for example, being able to know which products that can't interact and also producing a project plan

2.Data Understanding

In this phase, we start the initial data collection and familiarize ourselves with the data. This phase includes:

- Gathering initial data: In this step, we look for the data that we will need for our project. Data gathering could be done from different sources, for our case, data scraping for products and ingredients, and other sources like Kaggle.
- Describing data: It includes everything about data source, date, and number of columns and rows.
- Exploring data: It's like an initial analysis for the data where we use visualization to understand the content and the characteristics of the data.
- Verifying data quality: This ensures the data is accurate and reliable for our needs

3. Data Preparation

Data is rarely clean. This phase is dedicated to cleaning and transforming raw data into a suitable format for modeling.

4. Modeling

With clean data in hand, various modeling techniques are applied so in this phase we're going to focus on selecting modeling techniques ,tests, Building the model and Assessing it.

5. Evaluation

In this phase we will evaluate the performance of the model and ensure it meets the business objectives before the deployment by evaluating the results , reviewing the process and determining the next step.

6. Deployment

In this phase, we move the project from the development environment to the real world. This is where we integrate the model into a mobile app so users can actually use it. It includes making sure the model works smoothly within the app interface, setting up how the app sends data to the model, and providing a way to monitor the recommendations to ensure they stay accurate for the users.

Chapter 2:

Data Understanding and Preprocessing

I. Ingredient Compatibility Detection

This chapter presents the exploratory data analysis and preprocessing steps applied to the Ingredient Conflict Detection dataset. This is the part of the project focused on predicting whether two skincare ingredients are safe or risky to combine

2.1 Dataset Description

2.1.1 Data Source :

We used **ingredient_conflict_dataset.csv** which is composed of two datasets that are merged together before training:

1. Ingredient Feature Dataset (**ingredients_properties.csv**)

This dataset describes each ingredient using structured numerical and categorical properties. These features allow the model to understand how an ingredient behaves chemically and dermatologically without relying on ingredient names alone

Feature	Type	Description
ingredient	Text	Name of the ingredient (used as key for merging)

pH	Float	pH value of the ingredient (range: 2.5 to 7.0)
category	Categorical	Type of ingredient: AHA, BHA, retinoid, antioxidant, humectant, lipid, peptide, soothing, etc.
is_acid	Binary (0/1)	Whether the ingredient is an acid
is_oxidizer	Binary (0/1)	Whether the ingredient is an oxidizing agent
photosensitizing	Binary (0/1)	Whether the ingredient increases photosensitivity
exfoliant	Binary (0/1)	Whether the ingredient acts as an exfoliant
barrier_repair	Binary (0/1)	Whether the ingredient helps repair the skin barrier
antioxidant	Binary (0/1)	Whether the ingredient has antioxidant properties
humectant	Binary (0/1)	Whether the ingredient attracts moisture (humectant)
brightening	Binary (0/1)	Whether the ingredient has brightening effects
anti_inflammatory	Binary (0/1)	Whether the ingredient has anti-inflammatory properties
cell_turnover	Binary (0/1)	Whether the ingredient promotes cell turnover

The dataset contains **251 unique ingredients** across **22 categories** including retinoids, AHAs, BHAs, PHAs

2. Ingredient Pair Dataset (pairs_labels.csv)

The second dataset contains pairs of ingredients with a binary label indicating whether combining them is safe or risky. Each row represents one combination:

- **ingredient_a**: First ingredient name
- **ingredient_b**: Second ingredient name
- **risk**: Target variable => 0 = Safe, 1 = Risky (Incompatible)

The pairs were generated using a rule-based conflict engine with 15 chemical rules derived from dermatological knowledge (exp : retinoid + acid = conflict, oxidizer + antioxidant = conflict). The balanced version of the dataset contains **4372 pairs**

2.1.2 Dataset Overview :

After merging, the final dataset used for analysis has the following structure :

Property	Value	Details
Total rows (pairs)	4,371	Each row represents one combination of two ingredients

Total columns	33	ingredient_a, ingredient_b, risk + 15 features × 2 (one per ingredient)
Unique ingredients	94	
Target variable	risk (binary)	0 = Safe / Compatible, 1 = Conflict / Incompatible
Feature types	Numeric + Categorical	pH (float), category (text), 10 binary flags, irritation & strength levels (0-2)

Column Structure

Each row in the dataset represents a pair of ingredients A and B. For each ingredient the following features are available:

- pH : acidity level of the ingredient (float, range 3.0 to 7.0)
- Category : functional type: AHA, BHA, PHA, retinoid, antioxidant, humectant, lipid, peptide, brightener, soothing, etc.
- 10 binary flags (0 or 1): is_acid, is_oxidizer, photosensitizing, exfoliant, barrier_repair, anti_inflammatory, brightening, humectant, retinoid, antioxidant, comedogenic
- irritation_level : ordinal score: 0 = low, 1 = medium, 2 = high
- strength_level : ordinal score: 0 = mild, 1 = moderate, 2 = strong

2.2 Exploratory Data Analysis (EDA)

2.2.1 Univariate Analysis :

We use Univariate analysis to look at each variable individually to understand its distribution and basic statistics before looking at relationships between variables

Target Variable: Class Distribution

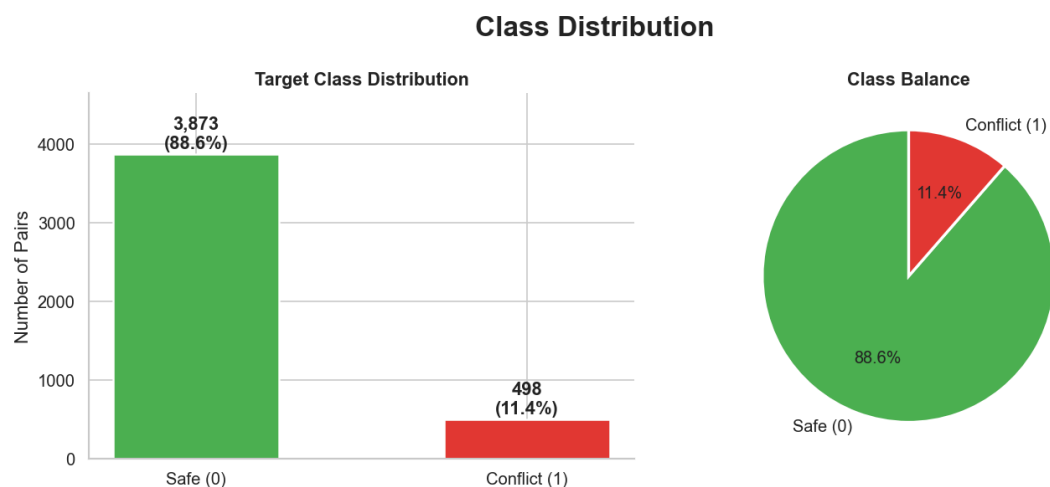


Figure 1 – Target class distribution:

Class	Count	Percentage
Safe (0)	3 873	88.6 %
Risky (1)	498	11.4 %

This result shows a significant class imbalance: only 11.4% of the pairs are conflicting which is realistic because most ingredient combinations are actually compatible

So we solved this imbalance before training by applying the SMOTE (Synthetic Minority Over-sampling Technique) on the training set only to avoid data leakage

pH Distribution

pH is one of the most important features in our dataset as we know that our skin has a natural pH between **4.5 and 5.5** so the pH of a product directly affects its safety and compatibility because values that are too acidic or too alkaline can cause irritation or reduce the ingredient effectiveness

Among the continuous numerical variables in our dataset, the most important are **pH_a** and **pH_b** which represent the acidity level of each of the two ingredients. These features help us understand the interaction between ingredients .

Their descriptive statistics are presented below to better understand their distribution

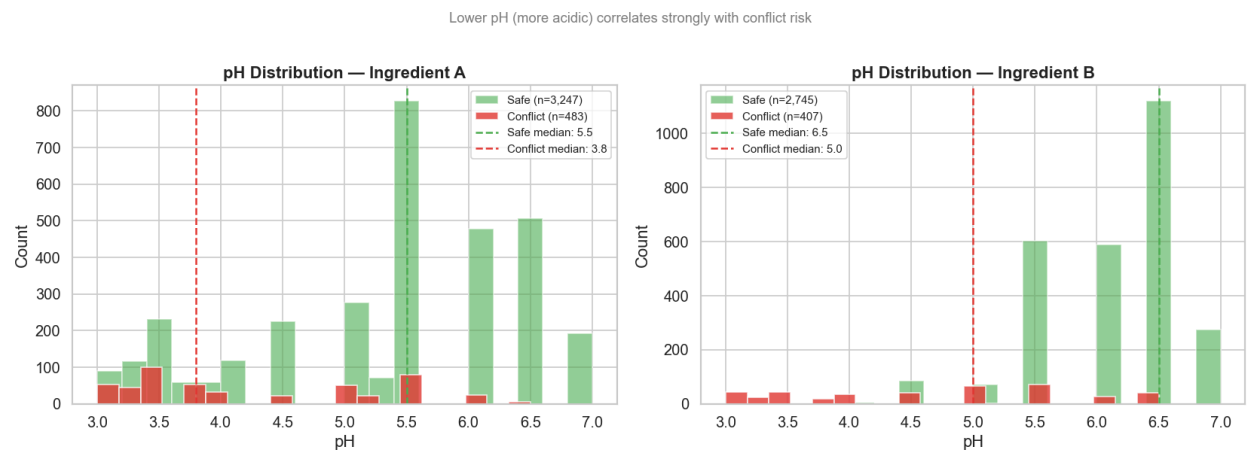


Figure 2 – pH distribution for ingredient A and B, split by safe vs conflict pairs.

Statistique	pH_a	pH_b	Interpretation
Count (non-nuls)	3 730	3 152	Ingredient B contains more missing pH values that ingredient A because of the merge gaps
Min	3.0	3.0	The minimum pH for both ingredients is 3.0 which means that some ingredients are strongly acidic
Max	7.0	7.0	The maximum pH for both was 7.0 which corresponds to neutral conditions
Médiane (Safe)	5.5	6.5	From the result we can deduce that safe combinations generally have higher pH values which means closer to the skin’s natural range
Médiane (Conflit)	3.8	5.0	We can observe that conflict pairs are clearly more acidic and that can explain their higher risk

Category Distribution

When we look at ingredient categories we can deduce that AHA is the most frequent category for ingredient B (with ~650 occurrences), followed by retinoid (~625) and brightening (~625). For ingredient A, lipid and humectant categories appear most often. The conflict rate line on the chart clearly shows that AHA, BHA, PHA, and retinoid categories have the highest conflict rates, while humectant, soothing, and protein categories have near-zero conflict rates.

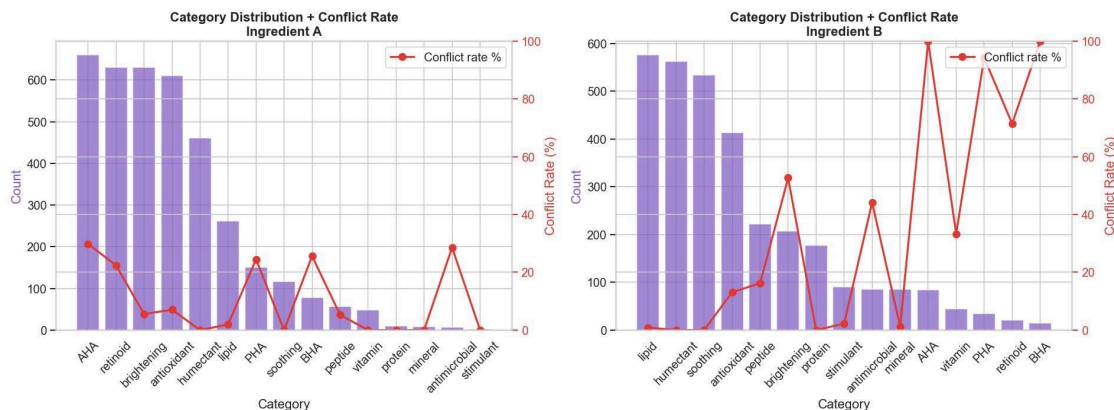


Figure 3 – Category distribution and conflict rate for ingredient A (left) and ingredient B (right)

Irritation & Strength Levels

The `irritation_level` and `strength_level` variables are ordinal features, which means that 0 is low, 1 is medium, and 2 is high. The plots show that most conflict pairs happen when the level of irritation is high and strong. On the other hand, most of the ingredients in safe pairs have an irritation level of 0 (low). This same pattern appears for both ingredient A and ingredient B, which means that these features are important and can help the model make good predictions.

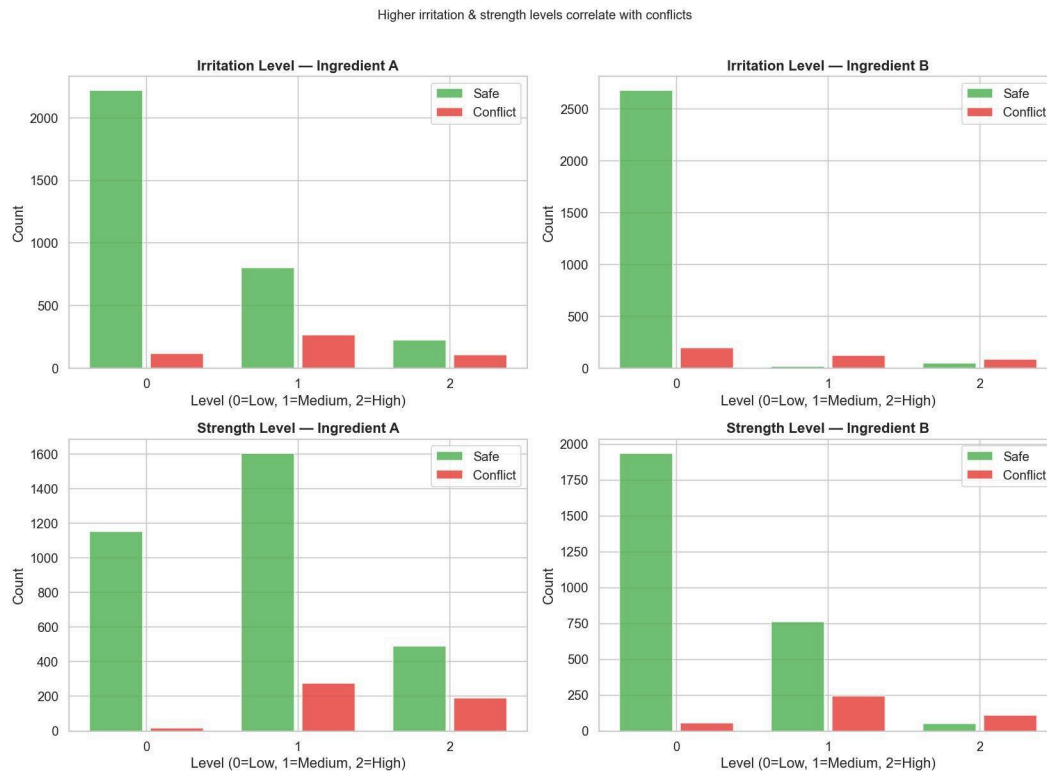


Figure 4 – Irritation and strength level distributions for ingredient A and B, split by safe vs conflict

2.1.4 Bivariate Analysis :

We use Bivariate analysis to study the relationship between each individual feature and the target variable (risk)

Binary Property Flags vs Conflict Rate

The chart below shows the conflict rate for ingredient B features when a property is present (= 1) vs absent (= 0). The dashed line marks the overall conflict rate of **11.4%**:

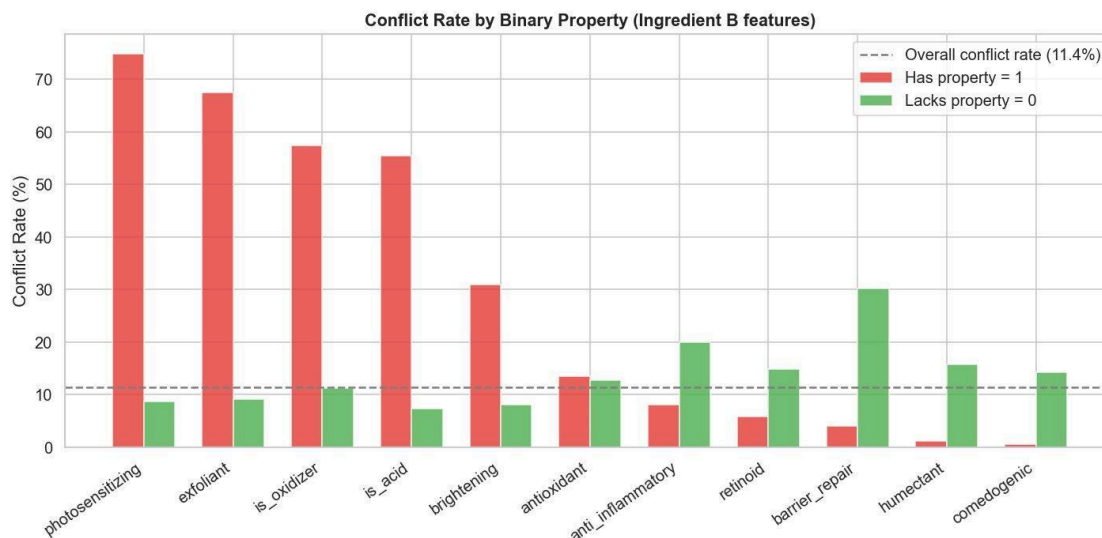


Figure 5 – Conflict rate by binary property flag for ingredient B. Properties above the dashed line (11.4%)

From this Chart can deduct that when :

- photosensitizing = 1 => conflict rate = ~75% (vs 9% when absent)
- exfoliant = 1 => conflict rate = ~68% (vs 9% when absent)
- is_oxidizer = 1 => conflict rate = ~58% (vs 11% when absent)
- is_acid = 1 => conflict rate = ~56% (vs 7% when absent)
- barrier_repair => 1 = conflict rate = ~4% (vs 30% when absent)
- humectant = 1 => conflict rate only= ~1% (vs ~16% when absent) it is very safe property

We can conclude that this analysis is one of the most informative in the EDA because it directly shows which chemical properties drive conflict risk

2.1.3 Multivariate Analysis :

We use Multivariate analysis to study the relationship between multiple features at the same time, as well as their combined effect on the target variable

Feature Correlation with Risk

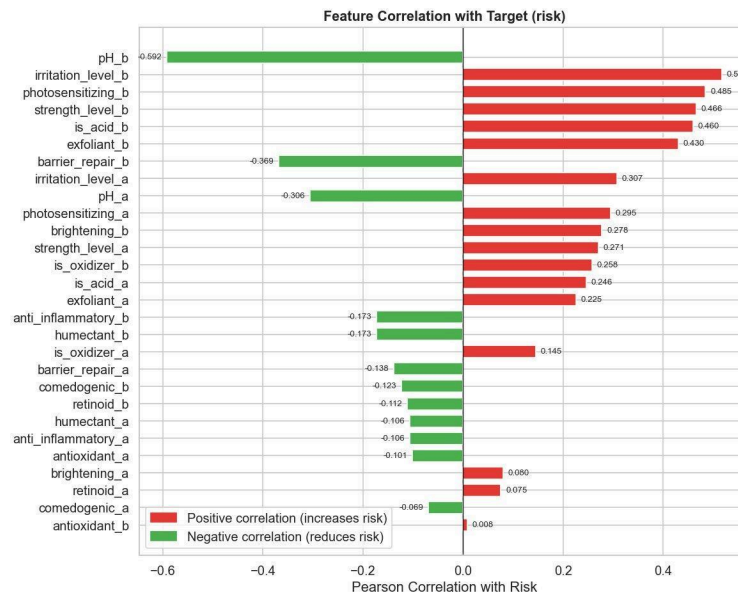


Figure 6 – Correlation of all features with risk

Features	Correlation	Interpretation
pH_b	-0.592	Our Strongest predictor is pH of ingredient B => higher pH_B = Safer Combination
irritation_level_b	+0.518	A high irritation level of ingredient B increases the risk of conflict especially when it is combined with other strong actives
photosensitizing_b	+0.485	Photosensitizing ingredients in B also increases the risk of conflict
strength_level_b	+0.466	When the strength level of an ingredient B is high it leads to more conflicts and that shows that intensity plays a role in compatibility
is_acid_b	+0.460	Acidic ingredients in B are more likely to be risky, especially in combinations
exfoliant_b	+0.430	Exfoliant B = frequent conflicts

Full Feature Correlation Matrix (lower triangle)

	pH_a	is_acid_a	is_oxidizer_a	photosensitizing_a	exfoliant_a	barrier_repair_a	anti_inflammatory_a	brightening_a	humectant_a	retinoid_a	antioxidant_a	comedogenic_a	irritation_level_a	strength_level_a	pH_b	is_acid_b	is_oxidizer_b	photosensitizing_b	exfoliant_b	barrier_repair_b	anti_inflammatory_b	brightening_b	humectant_b	retinoid_b	antioxidant_b	comedogenic_b	irritation_level_b	strength_level_b
pH_a	1.00																											
is_acid_a	-0.94	1.00																										
is_oxidizer_a	-0.24	0.17	1.00																									
photosensitizing_a	-0.10	0.04	-0.07	1.00																								
exfoliant_a	-0.11	0.03	-0.04	0.20	1.00																							
barrier_repair_a	0.04	-0.11	-0.06	0.09	0.05	1.00																						
anti_inflammatory_a	0.12	-0.05	-0.12	-0.37	0.03	0.21	1.00																					
brightening_a	-0.18	0.11	0.11	0.14	-0.13	-0.49	-0.12	1.00																				
humectant_a	0.08	-0.16	-0.06	-0.18	-0.05	-0.44	-0.14	-0.14	1.00																			
retinoid_a	0.19	-0.31	-0.08	0.06	0.25	-0.15	0.03	0.23	-0.07	1.00																		
antioxidant_a	0.13	-0.19	0.23	-0.28	-0.32	0.27	0.03	0.13	0.23	0.24	1.00																	
comedogenic_a	0.20	-0.00	-0.01	-0.15	-0.12	0.00	-0.15	0.21	-0.05	-0.12	0.16	1.00																
irritation_level_a	-0.16	0.23	0.23	0.16	0.41	-0.40	-0.40	0.27	0.16	0.16	-0.16	-0.16	1.00															
strength_level_a	-0.16	0.06	0.23	0.04	0.33	-0.49	-0.36	0.39	-0.41	0.20	0.12	-0.28	0.17	1.00														
pH_b	0.13	0.08	0.04	0.24	0.10	0.11	0.04	0.08	0.06	-0.17	0.15	0.04	-0.17	-0.17	1.00													
is_acid_b	-0.10	-0.05	-0.03	0.20	0.09	-0.04	-0.07	0.07	-0.01	0.13	-0.11	0.02	0.14	0.13	-0.10	1.00												
is_oxidizer_b	0.00	-0.01	-0.01	-0.02	0.01	-0.01	0.02	-0.02	-0.00	0.02	0.02	0.02	0.03	0.03	0.13	0.04	1.00											
photosensitizing_b	-0.02	-0.03	-0.03	0.16	-0.01	-0.01	-0.07	0.07	-0.04	0.17	-0.09	-0.01	0.11	0.12	-0.16	0.35	0.34	1.00										
exfoliant_b	-0.10	0.00	-0.01	0.23	0.07	-0.10	-0.09	0.08	-0.04	0.16	-0.10	-0.05	0.16	0.16	-0.11	0.02	-0.05	0.45	1.00									
barrier_repair_b	0.04	-0.10	0.01	-0.19	-0.11	0.12	0.04	-0.09	0.10	-0.14	0.12	0.06	-0.10	-0.16	-0.42	-0.14	-0.27	-0.35	-0.11	1.00								
anti_inflammatory_b	0.05	-0.02	-0.06	-0.04	-0.02	0.06	0.01	-0.05	0.05	0.04	0.04	-0.05	0.06	-0.10	0.13	-0.24	-0.22	-0.20	-0.14	-0.14	1.00							
brightening_b	-0.14	0.10	-0.																									

The correlation matrix shows expected relationships between features of the same ingredient. For example, **pH_a** and **is_acid_a** are strongly negatively correlated (-0.84), which makes sense because acidic ingredients naturally have lower pH values. Also, **irritation_level** and **strength_level** are highly positively correlated ($+0.73$) that means stronger ingredients tend to be more irritating. These types of correlations are normal in chemistry because many properties

are naturally linked so these relationships reflect real chemical behavior and they are not errors in the data

Top 15 Ingredients Most Involved in Conflicts

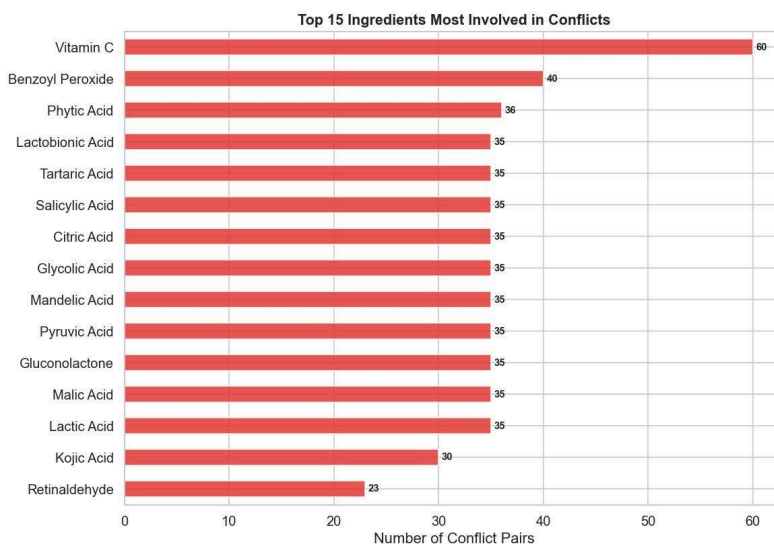


Figure 8 – Top 15 ingredients by number of conflict pairs they appear in

Category to Category Conflict Heatmap

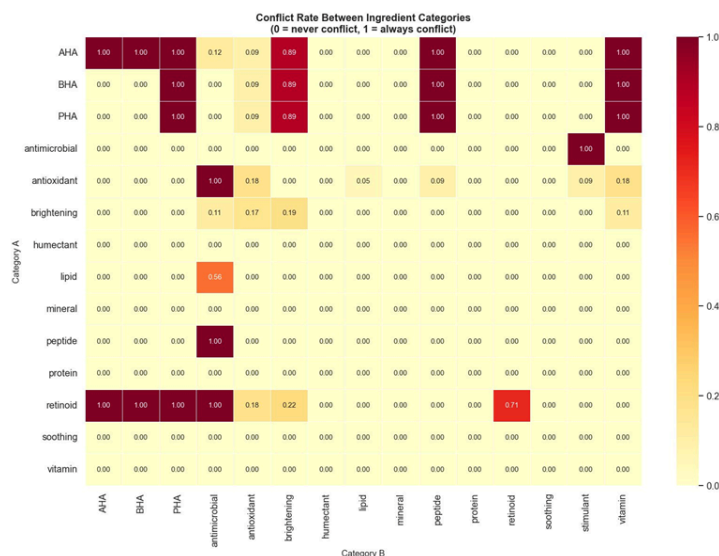


Figure 9 Conflict rate between category pairs (0 = never conflict, 1 = always conflict)

The heatmap shows a few clear patterns of how different categories of skincare ingredients interact together :

- AHA × AHA, AHA × BHA, and AHA × PHA = 1.0: Mixing any exfoliating acids always causes a problem and this is expected because stacking acids makes the skin more sensitive and destroys the skin barrier
- AHA × vitamin, AHA × peptide = 1.0: acids are incompatible with vitamins and peptides because the low pH can degrade vitamins stability and change the shape of peptides
- Retinoid × AHA, retinoid × BHA, and retinoid × PHA = 1.0: these are the most well known dangerous combinations in skin care. Using retinoids with exfoliating acids makes irritation and sensitivity much worse
- Retinoid x Retinoid = 0.71 : Combining multiple retinoids can also be a problem because it can make irritation and photosensitivity more probable even if there isn't always a direct chemical conflict
- humectant, soothing, lipid, protein categories = 0% conflict: There is no conflict between those categories because these groups are generally safe and work well with almost all other ingredients, so they make good supporting ingredients in formulations

pH Difference Analysis

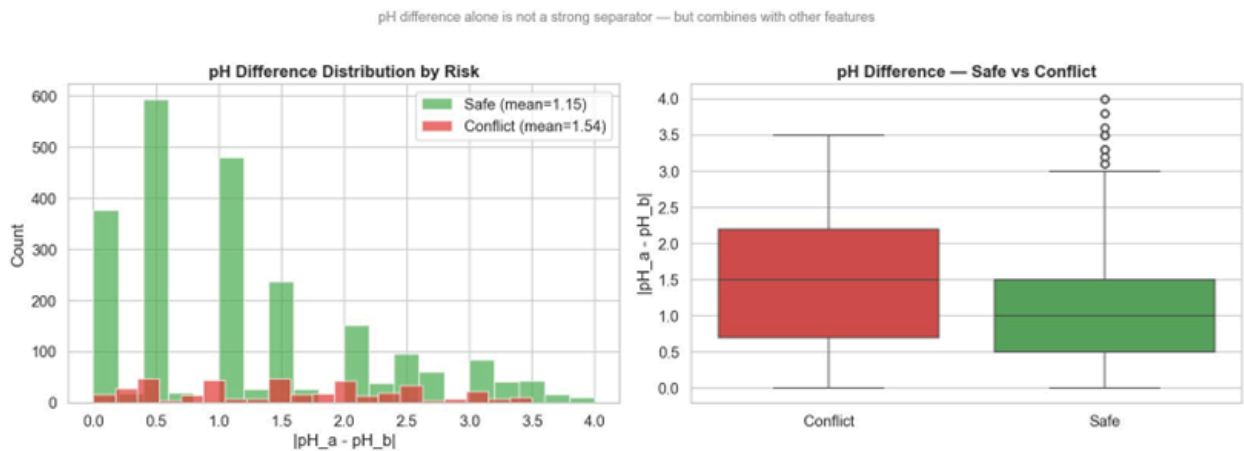


Figure 10 – pH difference distribution by risk class. Conflict mean = 1.54, Safe mean = 1.15.

From this figure we can observe that the average pH difference between conflict pairs is **1.54**, while the average pH difference between safe pairs is **1.15** , However pH difference alone isn't enough for a strong separator , it works best when it 's combined with other features . That's why we need to engineer pH_diff as an interaction feature instead of using pH_a and pH_b separately

2.3 Data Quality Assessment

Before applying preprocessing, we carefully evaluated the quality of our dataset to ensure that it becomes reliable and suitable for building meaningful analysis and models

Missing Values

We have a total of 30 out 33 columns containing missing values that are not random. They happen because some ingredient names in the pairs file didn't match any row in the ingredient properties file so all feature columns for that ingredient become null

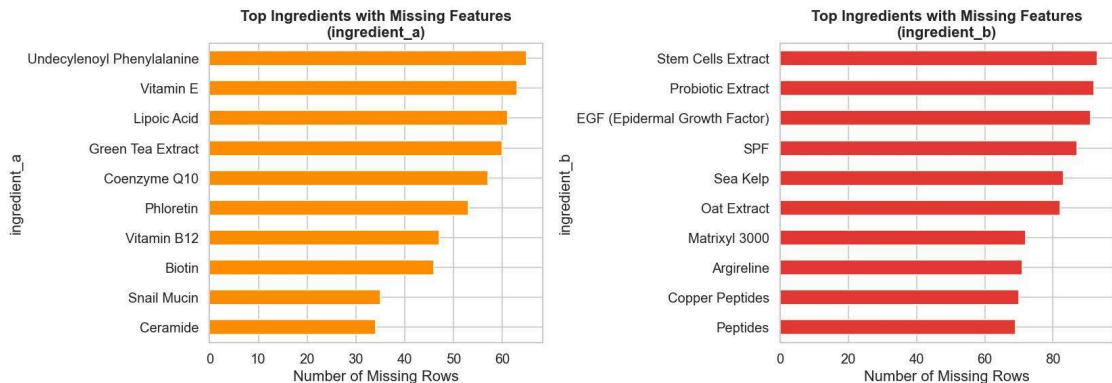


Figure 11 – Top ingredients with missing feature rows

Feature Group	Missing Count	Missing %	Cause
All B-side features (pH_b, category_b, etc.)	1,219	27.89%	Ingredient B not found in properties table for 1,219 pairs
All A-side features (pH_a, category_a, etc.)	641	14.66%	Ingredient A not found in properties table for 641 pairs
risk	0	0%	Target variable always present
ingredient_a, ingredient_b names	0	0%	Always present

Outlier Analysis

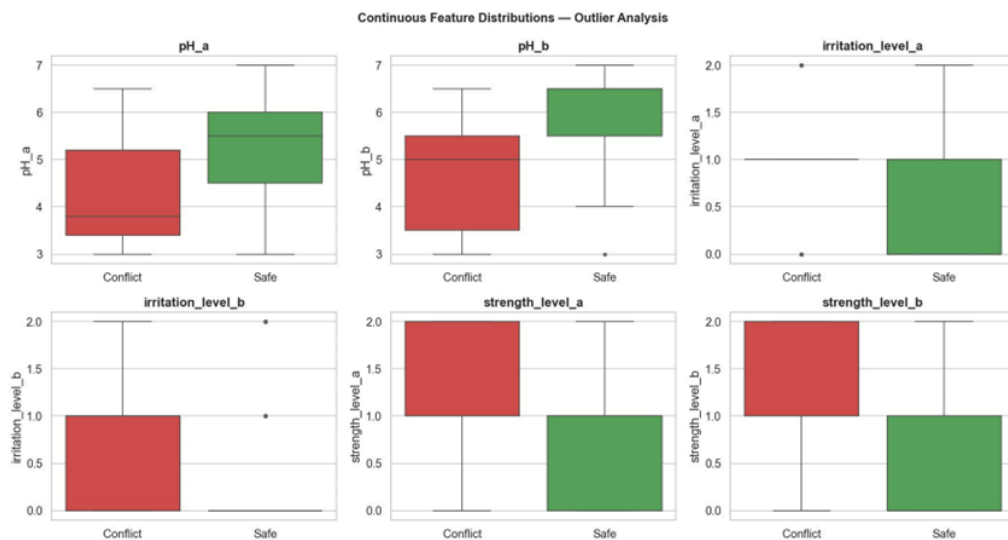


Figure 12 – Box plots of continuous features split by safe vs conflict class

To detect the outliers we used the **IQR method** (lower = $Q1 - 1.5 \times IQR$, upper = $Q3 + 1.5 \times IQR$), the outlier analysis revealed :

Feature	Outlier Count	Decision	Justification
pH_a	0 (0%)	Kept	There were no strange values found because all of the values were within the valid chemical range of [3.0,7.0]
pH_b	128 (2.9%)	Kept	These variations show real chemical diversity because some ingredients are naturally more acidic or closer to neutral
is_oxidizer_a	71	Kept	This is a binary feature, so applying IQR-based outlier detection is not appropriate
irritation_level / strength_level	Variable	Kept	These are ordinal features (0, 1, 2) and are naturally bounded, so no outlier treatment is needed
Binary flags (is_acid, humectant, etc.)	Variable	Kept	Same thing these are binary features so we don't need an outlier treatment

Conclusion: There were no outliers that were eliminated during the preprocessing. The values that the IQR method signaled in binary and ordinal columns were considered false positives, as this method assumes a continuous, normally distributed variable, which does not apply to this type of features.

The 128 flagged values in pHb were kept since they represent valid chemical variability and not errors and noise in the data

Dirty Data

The dataset doesn't contain dirty data like corrupted text, negative pH values or measurements that are physically impossible. The actual data quality issue was the mismatch between the names of ingredients in the pairs table and the properties table as ingredient names in the pairs table were not always equal to ingredient names in properties table that resulted in problems of join during merging and that resulted in missing values.

The real data quality problem was a name mismatch between the two source CSV files where ingredient names in the pairs table were not consistently equal to ingredient names in the properties table so the results of the mismatch created issues of join during merging

and that created missing values . This problem was fixed by recompiling both files of the same master ingredient list.

2.4 Data Preprocessing

Step 1 : Add Missing Indicators and Drop Ingredient Names

Before we dropped any columns , we made two binary flags to trace the missing ingredient information after the merge. These are **feature_a_missing** and **feature_b_missing** which show whether the ph of ingredient A or B is present or not (0=> value present , 1=> value missing) because missing ph values are not random they often mean that the ingredient is rare , oily

Then we dropped ingredient(a) and ingredient(b) since we cannot directly use raw ingredient names such as Retinol or Vitamin C in ML models

Result : The data now contains 29 numeric columns and 2 additional indicators flags of missing data and 2 category columns (total = 33 columns)

Step 2 : Encode Categorical Features

Unlike the ingredient name columns we kept the categorical columns because they include the functional category of every ingredient like AHA , BHA ect and also they have real and powerful predictive information

We use **LabelEncoder** to convert each ingredient into a format that the machine learning model can understand ,this was done by encoding each unique category into a consistent integer value and make sure that our data preserve its category identity

Step 3 : Impute Missing Numeric Values

We handled missing numeric values using **IterativeImputer with a Random Forest Regressor (max_itr=10)** we applied this method because it predicts the missing values using the relationships between all other features and also it captures non linear chemical relationships better than simple Imputer

Result : Missing values after imputation = 0

Step 4 : Feature Engineering (11 new features)

With the help of the EDA, we created 11 new features to better understand how two ingredients interact with each other. This step was necessary because the model cannot learn the interaction between two ingredients just by looking at their individual properties separately. For example, knowing that ingredient A is an acid and ingredient B is an acid

does not automatically tell the model that combining two acids is dangerous; we need to explicitly create a feature that captures this situation.

Each feature was inspired by a specific EDA finding and validated by dermatological literature.

- **pH_diff = |pH_a - pH_b|** : Each active ingredient works within a specific pH range. When two ingredients have very different pH values, the first disrupts the skin environment needed by the second. Our EDA showed conflict pairs have a mean pH_diff of 1.54 vs 1.15 for safe pairs. (*Draelos, 2016 — Cosmetic Dermatology*)
- **combined_irritation = irritation_a + irritation_b** (corr: +0.50) Skin irritation is cumulative. Two moderately irritating ingredients together exceed the skin barrier's recovery capacity. This is a well-documented phenomenon called cumulative irritation. (*Charbonnier et al., Contact Dermatitis, 1998*)
- **combined_strength = strength_a + strength_b** (corr: +0.511) Using two strong actives simultaneously overwhelms the skin before it can adapt. Dermatologists consistently advise introducing one strong active at a time. (*Mukherjee et al., Clinical Interventions in Aging, 2006*)
- **both_retinoid = retinoid_a × retinoid_b** All retinoids increase photosensitivity. Two retinoids together compound this risk without adding any therapeutic benefit. Our heatmap showed retinoid × retinoid has a 0.71 conflict rate. (*Gilchrest, British Journal of Dermatology, 1992*)
- **both_exfoliant = exfoliant_a × exfoliant_b** (corr: +0.251) Two exfoliants remove too much of the protective stratum corneum at once, causing over-exfoliation. Our heatmap showed AHA × AHA and AHA × BHA always conflict (rate = 1.0). (*Elias & Feingold, Skin Barrier, 2006*)
- **both_acid = is_acid_a × is_acid_b** (corr: +0.276) Two acids drop the skin's surface pH far below its natural range (4.5–5.5), disrupting the enzymes that maintain the barrier. Our EDA showed is_acid raises conflict rate from 7% to 56%. (*Schmid-Wendtner & Korting, Skin Pharmacology, 2006*)
- **oxidizer_count = is_oxidizer_a + is_oxidizer_b** (corr: +0.280) Oxidizers like benzoyl peroxide chemically destroy antioxidants and peptides through oxidation reactions. We use addition (not multiplication) because even one oxidizer in the pair already creates risk. (*Pinnell et al., Dermatologic Surgery, 2001*)
- **photo_sensitive_pair = photosensitizing_a × photosensitizing_b** (corr: +0.436) When both ingredients increase UV sensitivity, the combined photosensitivity risk becomes severe, especially dangerous in morning routines. Our EDA showed photosensitizing = 1 raises the conflict rate to 75%. (*Kaidbey et al., Photodermatology, 2003*)
- **strong_irritant_combo = strength_a × irritation_b + strength_b × irritation_a** Our most important feature. A strong ingredient compromises the skin barrier, making the skin more vulnerable to the irritant effects of the partner ingredient in both directions. This cross-product captures synergistic irritation, which is stronger than the sum of individual effects. This became the most important feature in our final model (importance = 0.133). (*McFadden & Basketter, Contact Dermatitis, 2000*)

- **acid_exfoliant_interaction = is_acid_a × exfoliant_b + is_acid_b × exfoliant_a** (corr: +0.286) An acid lowers the skin's pH, which amplifies the activity of any exfoliant in the same routine beyond its intended concentration. (Yu & Van Scott, *Cosmetic Dermatology*, 1994)
- **pH_risk = max(|pH_a - 5|, |pH_b - 5|)** We hypothesized that ingredients deviating far from optimal skin pH (5.0) would be riskier. The correlation was weak (−0.044), likely because the direction of deviation matters more than the distance. We kept the feature since it did not harm model performance. (Ali & Yosipovitch, *Acta Dermato-Venereologica*, 2013)



Figure 13 – Engineered Features Distribution by Risk Class

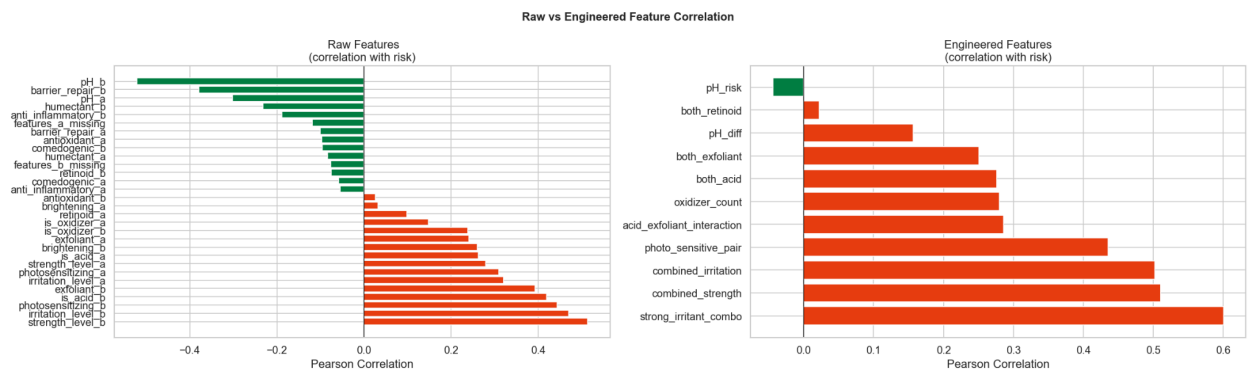


Figure 14 – Correlation of raw vs engineered features with risk

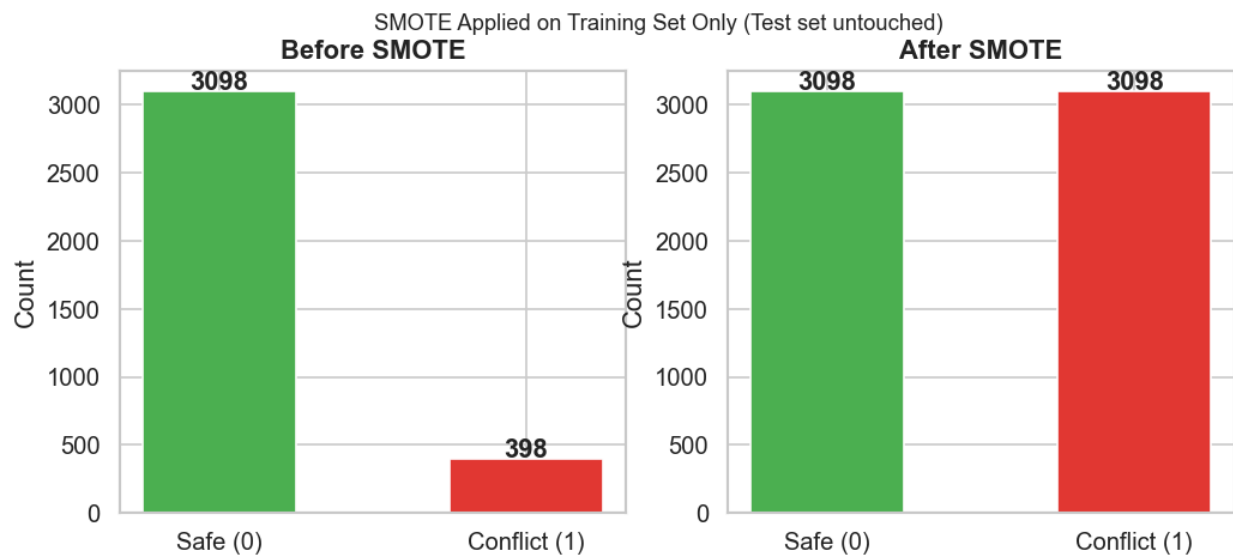
Step 5 : Train / Test Split

We splitted the dataset into training (80%) and testing (20%) sets and we used stratification to preserve the original class distribution

Set	Rows	Safe (0)	Conflict (1)
Training set (80%)	3,496	3,098 (88.6%)	398 (11.4%)
Test set (20%)	875	775 (88.6%)	100 (11.4%)

Step 5 : Handle Class Imbalance with SMOTE

Our target class is highly imbalance (ratio = 7.78) so to handle it we applied SMOTE (Synthetic Minority Over-sampling Technique) on the training set only to avoid data leakage



	Before SMOTE	After SMOTE
Safe (0)	3,098 samples (88.6%)	3,098 samples (50.0%)
Conflict (1)	398 samples (11.4%)	3,098 samples (50.0%)
Total	3,496	6,196

II. Skin Compatibility (Type & Concern) Prediction

This chapter presents the exploratory data analysis and preprocessing applied to the skincare dataset. This part of the project focuses on determining the skin type and skin concern targeted by a product using the list of ingredients of this product.

2.1 Dataset Description

2.1.1 Data Source :

At the beginning of this project, a synthetic dataset was used, consisting of a merge of two Kaggle datasets into one artificially constructed dataset. However, this dataset showed inconsistencies and models were not learning patterns but memorizing them. With further research, a better quality dataset was found. The final selected dataset is Kaggle 19000+ Skincare Products Database of Skinsort a dataset with initial features described in the table below:

Feature	Type	Description
brand	text	the brand of the product
name	text	the name of the product
type	Categorical	Type of product: serum,moisturizer, mask etc...
country	text	the origin of the product
ingredient	text	list of ingredients of the product

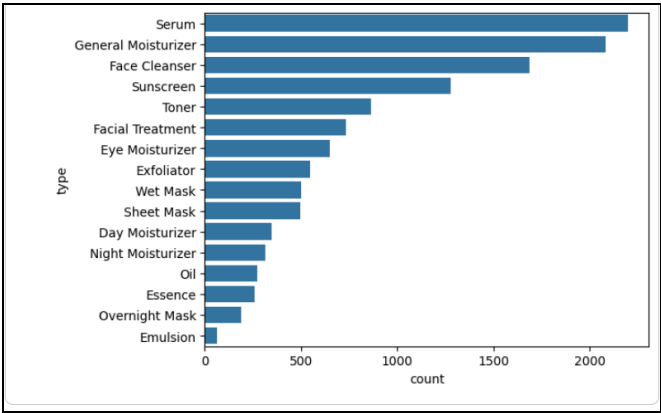
afterUse	text	The text contains different usage effects, including the concerns addressed by the products and the skin types they are suitable or unsuitable for. By mapping this column, we were able to perform feature engineering and create the target variables.
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2.1.1 Dataset Overview :

Name	Kaggle:19000+ Skincare Products Database of Skinsort
Description	Comprehensive dataset consisting of over 19,000 product details extracted from the Skinsort website.
Initial Size	Rows :19,050 ,Columns : 6
Size	9,510 products with 10 columns,product type OHE matrix (16 types), TF-IDF ingredient matrix (500 features)
Columns	• product_type OHE: 9,510 × 16 • afterUse: column containing a list of after-use effects and skin instructions. The tags extracted from this column were mapped into: → skin_type 5 binary columns (Dry, Oily, Normal, Combination) → skin_concern : 6 classes (Acne,Dryness, Eczema, Hyperpigmentation, Redness, Sensitivity) → age_group:5 classes (18-25, 26-35, 36-45, 46-55, 56+) • TF-IDF ingredient matrix: 9,510 × 500 • brand, country, name → dropped
Quality	• 19,050 raw products → 9,510 after filtering to face skincare and removing nulls.

	Labels derived from user reports about product effects
Distribution	- skin type label rate → moderately imbalanced. - skin_concern → moderately imbalanced - age label rate→Mild imbalance

2.2 Exploratory Data Analysis (EDA)



2.2.1 Product Type Distribution

Serums and General Moisturizers dominate the dataset, while Emulsions and Overnight Masks are the least represented. This distribution reflects real market trends but also signals that the dataset is unbalanced across product types, however after more EDA we decided not to use product types due to its weak correlation with our targets.

Figure 16 – Product Type Distribution

2.2.2 AfterUse Tags Frequency

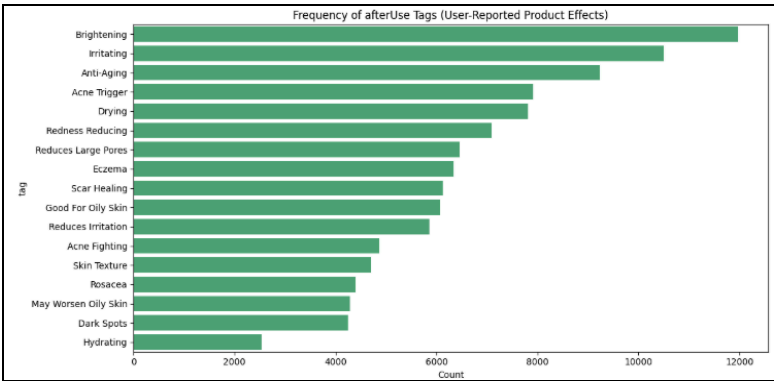


Figure 16 – AfterUse Tags Distribution

Brightening and Irritating are the most reported user effects, appearing in over 10,000 and 8,000 products respectively. Tags like Hydrating and Dark Spots are the rarest. This confirms that the afterUse column is rich in signal but heavily skewed toward certain effects, which directly impacted the label engineering process — rare tags produce underrepresented labels and contribute to class imbalance in the targets.

2.2.3 Skin Type Distribution

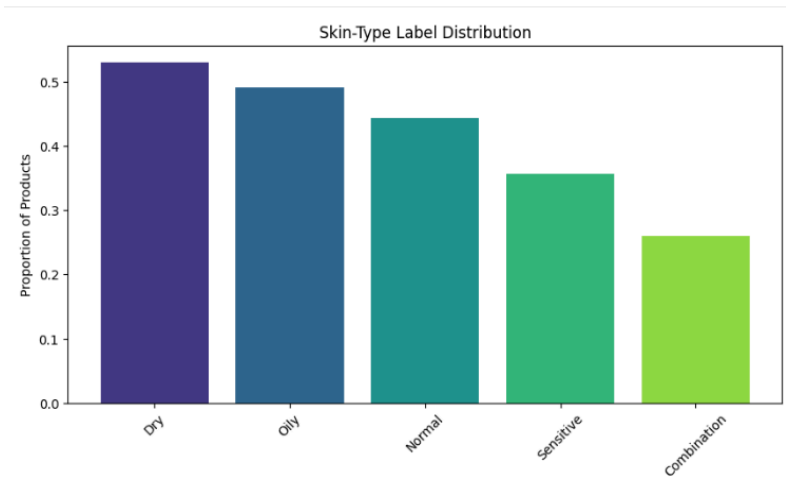


Figure 17 – Skin Type Distribution

Dry has the highest proportion followed by Oily and Normal , while Combination is the least represented.The distribution is moderately imbalanced but acceptable for modeling. This confirms that no single skin type dominates overwhelmingly, and `class_weight='balanced'` is sufficient to handle the imbalance without resampling.

2.2.4 Skin concern Distribution

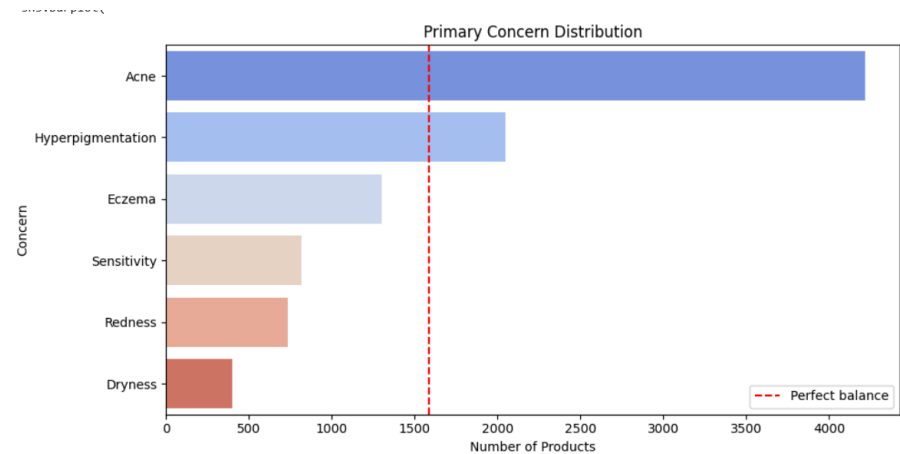


Figure 18 – Skin Concern Distribution

Acne is the dominant followed by Hyperpigmentation and Eczema. Dryness is the rarest class at under 500 products. This severe imbalance where Acne alone represents over 44% of the dataset, justifies the use of `class_weight='balanced'` and makes macro F1 a critical evaluation metric to ensure minority classes are not ignored during training.

2.2.5 Products per Skin Type + Labels per Product

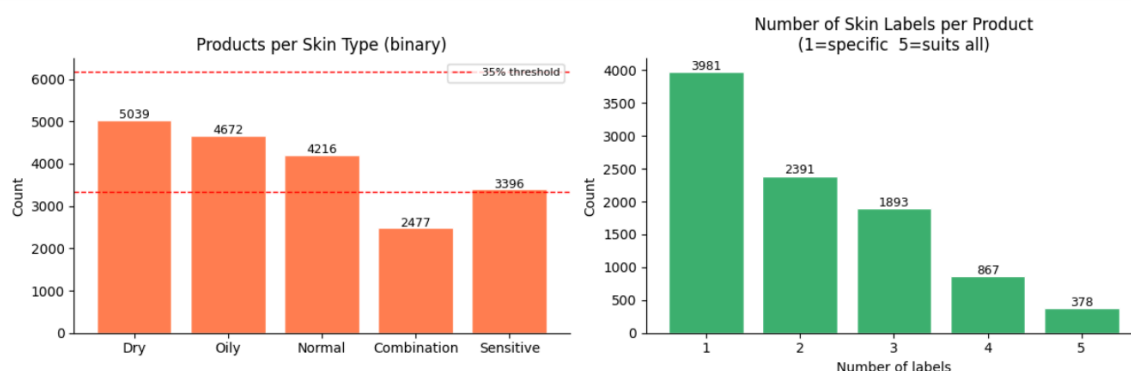


Figure 19 – Products per Skin Type + Labels per Product

The left chart confirms the binary label rates seen in the previous plot. The right chart shows that 3,981 products carry exactly one skin type label, making them the most specific and most reliable training examples. However, a large portion of products carry 3 or more labels, which makes the task harder since the model must learn to distinguish overlapping skin types from ingredients alone.

2.2.6 Ingredient Vocabulary Analysis

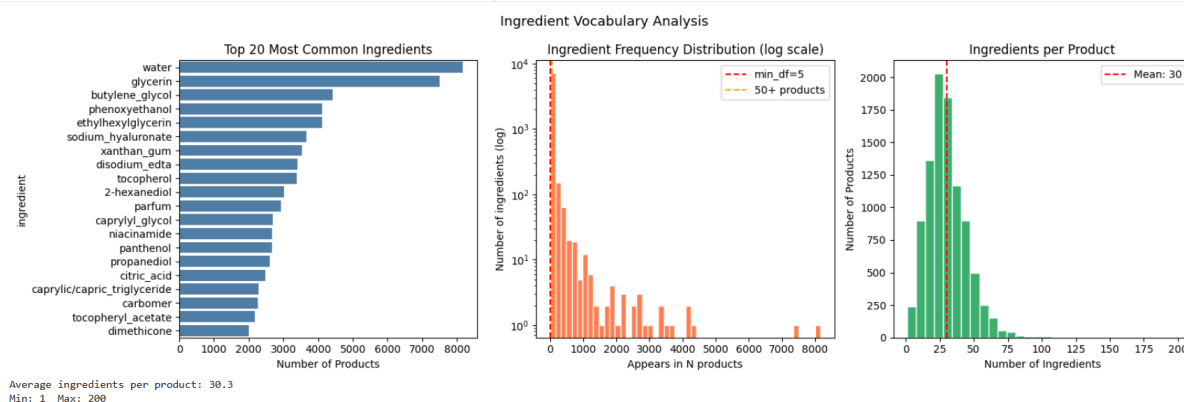


Figure 20 – Ingredient Vocabulary Analysis

Water, Glycerin, and Butylene Glycol are the most ubiquitous ingredients, appearing in over 7,000 products. The frequency distribution (log scale) confirms a heavily skewed vocabulary, most ingredients appear in very few products. This justifies the use of TF-IDF with `min_df=5` to filter out rare ingredients and `sublinear_tf=True` to prevent universal ingredients like Water from dominating the feature space.

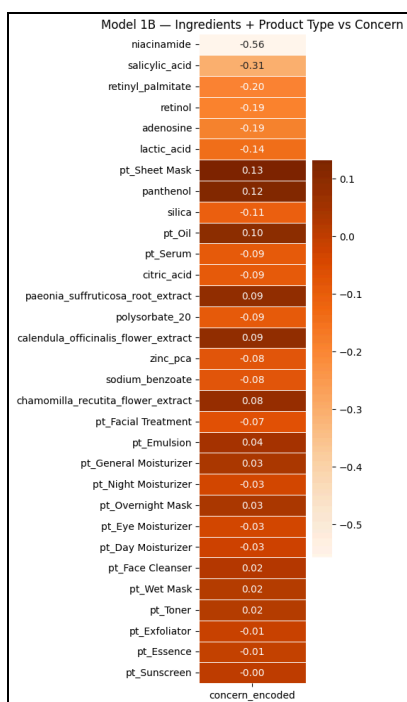
2.2.7 Model 1A: Top Ingredients vs Skin Types



Figure 21 – Correlation Top Ingredients vs Skin Types

Correlations are generally weak, with Panthenol showing the strongest positive correlation with Sensitive (0.57) and Niacinamide with Oily (0.57). Most ingredients show low correlations across all skin types, confirming that no single ingredient is a strong discriminative signal on its own. The model must learn from combinations of ingredients rather than individual ones, which justifies the use of ensemble methods like Random Forest over simpler linear approaches.

2.2.8 Model 1B: Ingredients + Product Type vs Concern



Niacinamide and Salicylic Acid showed the strongest correlation with concern prediction, mainly toward Acne. However, the correlations remained generally weak across all features, including product type variables, which mostly had near-zero correlation. Because of this, ingredients alone were tested in Model 1B to check whether ingredient information is sufficient for predicting concerns without depending on product type, since product type may introduce weak or circular signals.

Figure 22 – Correlation Top Ingredients & Product Type vs Skin Concern

2.3 Data Preprocessing

For the preprocessing step, we made sure to perform exploratory data analysis throughout this phase, which helped with better understanding for the data and preprocessing decisions steps:

Step 1: Filtering face products

In this step, we only kept the products that are used for face skincare based on domain knowledge and removed the ones that contain null values. After this filtering, we got 12,495 face products. As a result, we got 16 product types, with serum and moisturizer being the most dominant type. Generally, product types are imbalanced; the imbalance reflects real-world product availability as serums and moisturizers are most likely to exist and be used for skincare.

Step 2 : Mapping text tags into skin-type labels

In this step, we use the afterUse column to create new skin-type columns. This is done by mapping tags to skin types based on general skincare knowledge and explicit tag meanings. For example, “Good For Oily Skin” indicates that a product is suitable for oily skin, while “Drying” means the product is not suitable for dry skin, so it is assigned a value of 0. Another example is combination skin type representing oily and normal signals together, Irritation indicating sensitive skin type.

Distribution of skin types :

Moderate imbalance distribution as Dry and Oily dominate, while (Combination underrepresented) , most products carry 1 to 2 labels, meaning the model will have specific, meaningful targets to learn from rather than noisy overlapping signals.

Step 3:Text preprocessing for ingredients:

Each product's contents were listed in different formats; multi-word ingredients were separated incorrectly, and there was also noise, e.g., rare and misspelled ingredients.

To fix these problems ingredients text went through these steps below:

- splitting the text by commas so that each ingredient becomes token
- converting all text to lowercase for consistency
- merging multi-word ingredients into a single token (e.g., salicylic acid → salicylic_acid).
- TF-IDF was used to convert the text into a numerical matrix.

HyperParameters used :

Parametres	Values	Justification
max_features	500	We found 7,525 unique ingredients, but only 3,338 appear in more than 5 products. This means that many ingredients are rare and not useful for learning patterns. Therefore we used min_df = 5 to reduce noise
min_df	5	From the 3,338 ingredients, many were still not very informative, so we further reduced the feature space by setting max_features = 500.
ngram_range	(1,1)	Controls token size,Each word (token) is taken alone
strip_accents	unicode	Remove Accents ,text

		becomes standardized
sublinear_tf	True	prevent very frequent ingredients like water from dominating the model

Results:

The outcome is a 2D sparse (9510 × 500) matrix with values representing TF-IDF scores of that ingredient for that product, rows representing products, and columns representing ingredients.

Step 4:Encoding

Label encoding was applied to the concern target variable, converting categorical class names (Acne, Dryness, Eczema, Hyperpigmentation, Redness, Sensitivity) into integer values (0–5) following alphabetical order. No encoding was needed for the skin type targets since they are already stored as binary columns (0/1).

Step 5 : Train / Test Split

The dataset was split into 80% training and 20% testing using stratified sampling to preserve class distributions across both sets. The same split was applied to both models to ensure consistency in evaluation.

Set	Rows	Features
Training set (80%)	7608	500
Test set (20%)	1902	500

Chapter 3:

Modeling & Evaluation

I. Ingredient Compatibility Detection

This chapter describes how we built and compared models of the Ingredient Conflict Detection task. We trained four classification models, evaluated their performance, and chose the best, then we tuned and cross-validated it.

3.1 Why Classification?

Our target variable “**risk**” is binary (0= safe , 1=conflict) so this is a binary classification model , our objective is to predict if the interaction of two ingredients is safe or not .

We tested four models : Logistic Regression, Decision Tree, Random Forest, and XGBoost

3.2 Evaluation Metric Choice

In our project we want to decrease the value of the false negative (when the model predicts “safe” but the pair is actually a conflict) This type of error is dangerous because it can lead to skin irritation or damage. That’s why we chose **Recall** (of the conflict class) as our most important metric since increasing Recall means fewer false negatives and better detection of real conflicts

Also we chose other metrics :

- **F1-Score:** This metric balances Recall and Precision. It also makes sure that the model does not over predict false conflicts and at the same time identifies most real ones
- **ROC-AUC:** This measure assesses the ability of the model to distinguish between pairs of safe and conflict using probability. It is primarily applied in the comparison of models that have similar performance

Note : we ignored Accuracy because our data is imbalanced

3.3 Models Trained

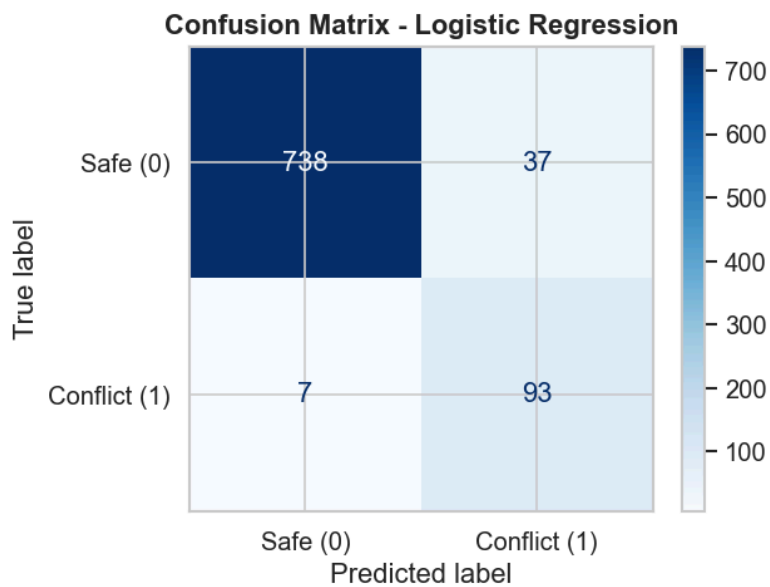
Model 1 : Logistic Regression

We used it as a simple baseline to measure how well a simple approach performs before trying more complex models. It was trained on scaled SMOTE data with `class_weight='balanced'`

Class	Precision	Recall	F1	AUC
Safe (0)	0.99	0.95	0.97	
Conflict (1)	0.72	0.94	0.82	0.9863

Final results :

- **Recall:** 0.9300
- **F1-Score:** 0.8087
- **ROC-AUC:** 0.9853



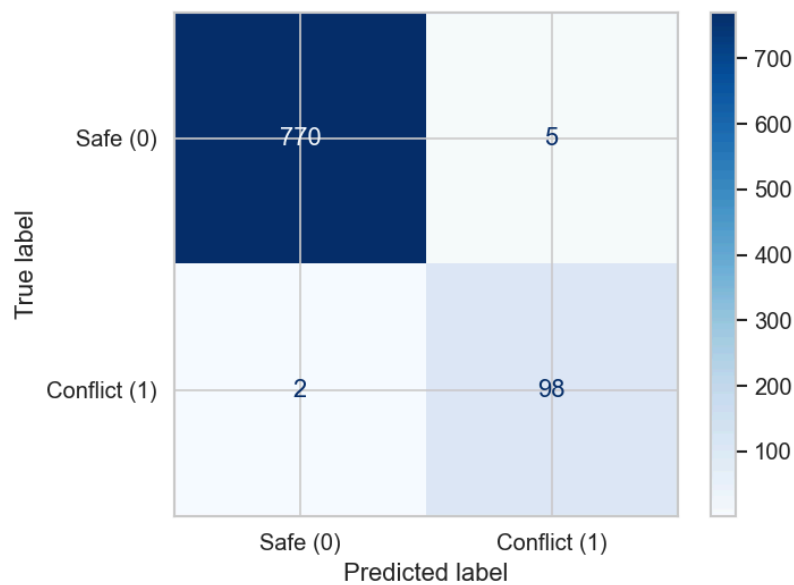
Model 2 : Decision Tree

We used it T with max_depth=10 and class_weight='balanced' and also it was trained on SMOTE data. Decision trees are interpretable and can capture non-linear patterns but tend to overfit on complex data

Class	Precision	Recall	F1	AUC
Safe (0)	0.99	0.98	0.99	
Conflict (1)	0.85	0.94	0.90	0.9875

Final results :

- **Recall:** 0.9800
- **F1-Score:** 0.9655
- **ROC-AUC:** 0.9875



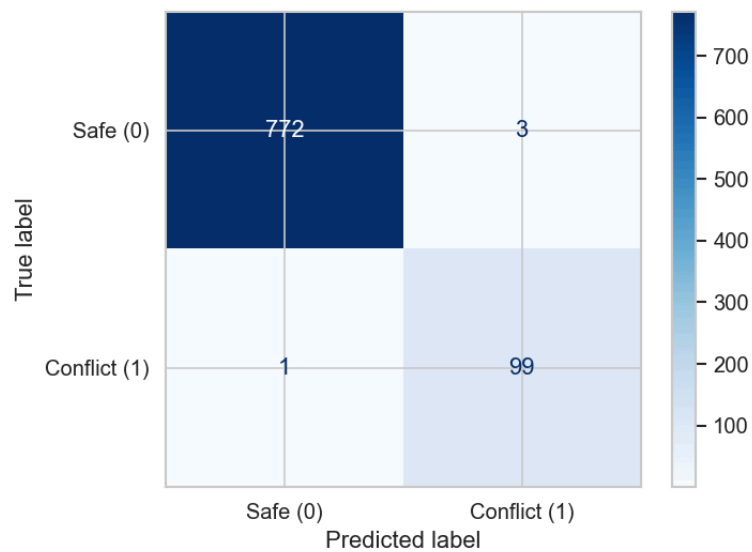
Model 3 : Random Forest

Random Forest is an ensemble of decision trees, it averages the predictions of 100 trees to reduce variance and improve generalization. We trained it with `n_estimators=100`, `class_weight='balanced'` and on SMOTE data

Class	Precision	Recall	F1	AUC
Safe (0)	1.00	1.00	1.00	
Conflict (1)	0.97	0.99	0.98	0.9933

Final results :

- **Recall:** 0.9900
- **F1-Score:**0.9802
- **ROC-AUC:** 0.9933



Model 4 : XGBoost

XGBoost is a gradient boosting algorithm that builds trees sequentially, each correcting the errors of the previous one. It was trained with $n_estimators=100$ on SMOTE data

Class	Precision	Recall	F1	AUC
Safe (0)	1.00	1.00	1.00	
Conflict (1)	0.98	0.97	0.97	0.9929

Final results :

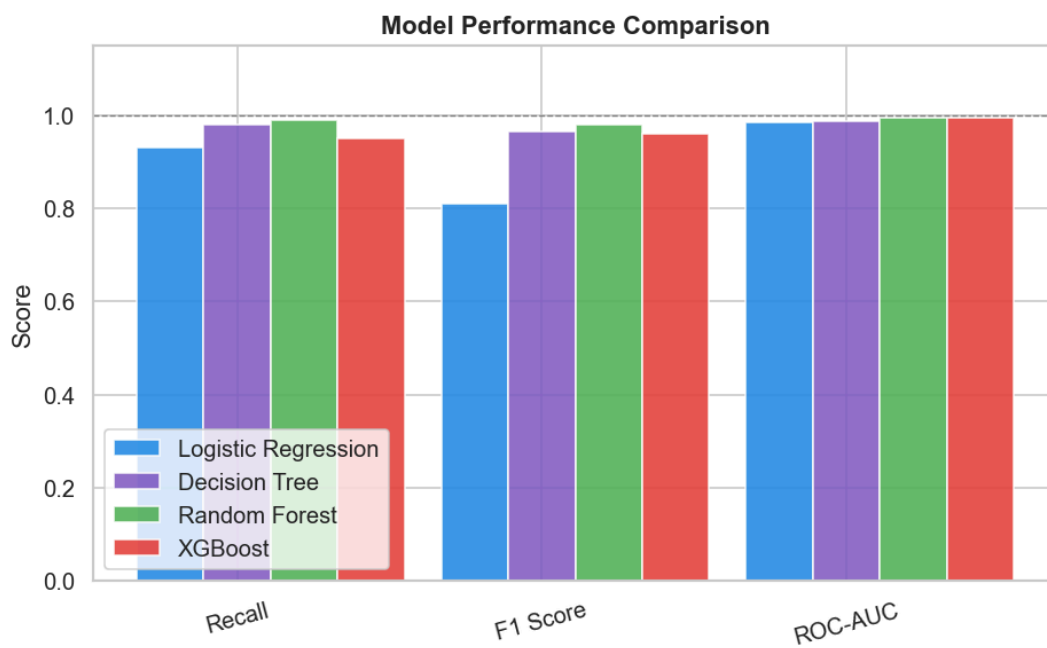
- **Recall:** 0.9500
- **F1-Score:** 0.9596
- **ROC-AUC:** 0.9929

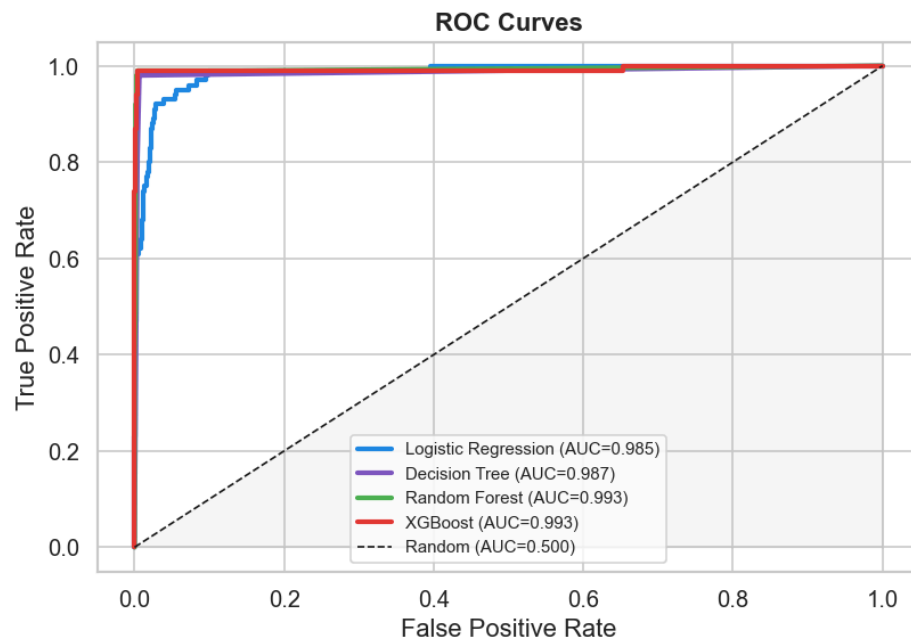
3.4 Model Comparison

Model	Recall	F1	ROC-AUC
Logistic Regression	0.93	0.8	0.98

Decision Tree	0.98	0.96	0.98
Random Forest	0.99	0.98	0.99
XGBoost	0.95	0.95	0.99

When we applied the selection criteria (step 1 : highest Recall , Step 2 : Highest F1 and Step 3 : ROC-AUC) we deduce that **Random Forest** is our best Model





3.5 Hyperparameter Tuning

We used GridSearchCV with 5-fold stratified K-fold cross-validation to optimize hyperparameters of our best model. The results are shown in the table below :

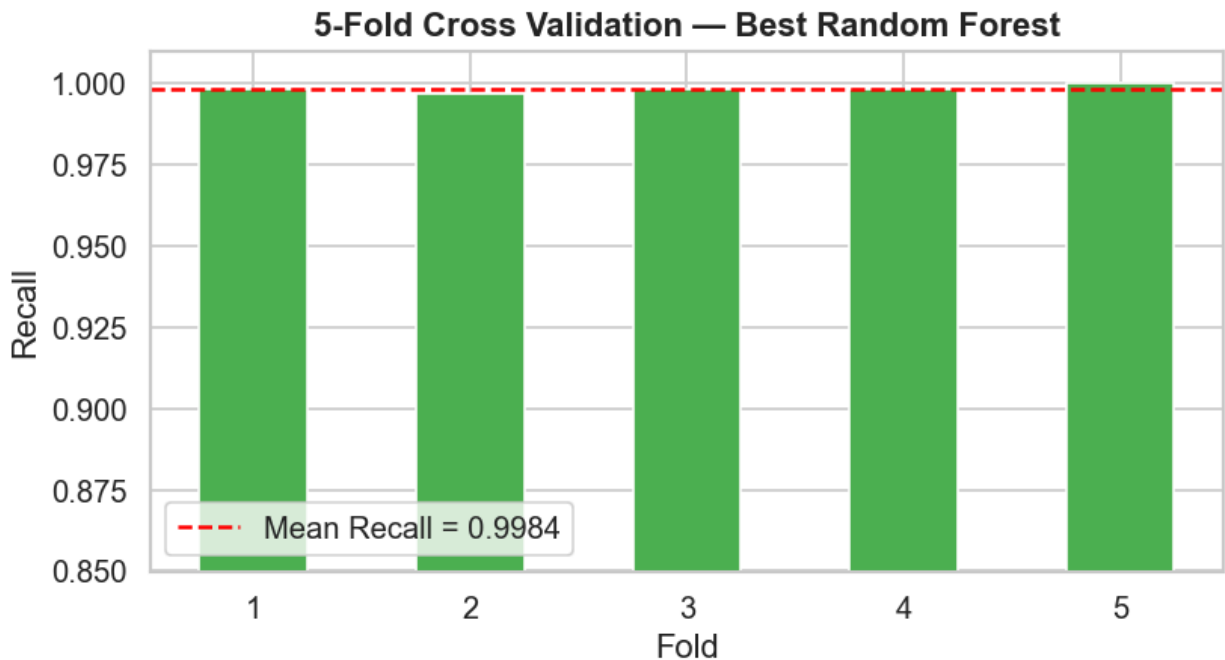
Hyperparameter	Values Searched	Best Value
n_estimators	[100, 200, 300]	200
max_depth	[None, 10, 20]	None
min_samples_split	[2, 5]	2
min_samples_leaf	[1, 2]	1

Tuned Model Results on Test Set

Metric	Before Tuning	After Tuning	Change
Recall	0.9900	0.9900	Stable
F1-Score	0.9802	0.9832	+0.003
ROC-AUC	0.9933	0.9947	+0.0014

3.6 Cross Validation

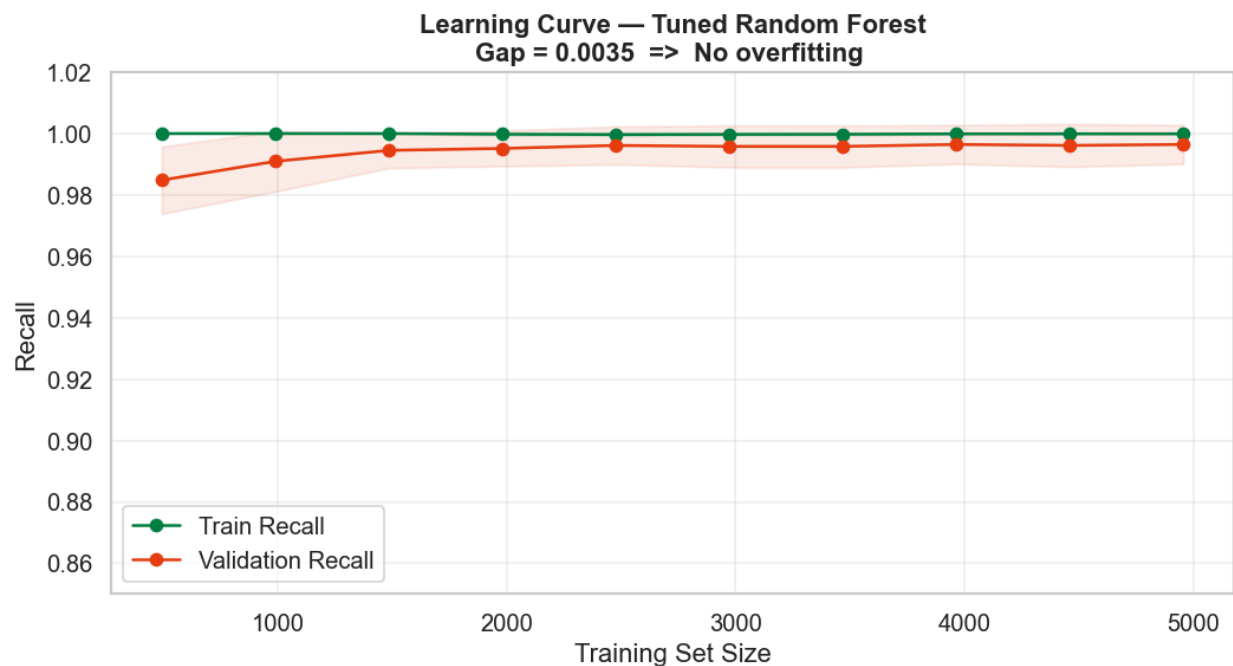
To ensure stability of the tuned Random Forest model, we applied the 5-fold Stratified Cross-Validation on the training set after applying SMOTE . The model achieved consistently high recall scores across all folds, with a mean recall of 0.9984. The very small variation between folds indicates excellent stability, robustness, and consistent performance of the model.



Overfitting check : Gap (CV Recall – Test Recall)

The learning curve shows that the tuned Random Forest model performs very well on both the training and validation data. The training recall remains close to 1.00, while the validation recall increases gradually and becomes stable around 0.996.

The small difference between the two curves (Gap = 0.0035) indicates that there is no overfitting. This means that the model generalizes well and gives consistent results on unseen data.



II. Skin Compatibility (Type & Concern)

Prediction

Two prediction tasks were defined from the same dataset. Model 1A predicts which skin types a product is suitable for, formulated as a multilabel problem where a product can belong to multiple skin type classes simultaneously. Model 1B predicts the primary skin concern a product addresses, formulated as a multiclass problem where each product is assigned exactly one concern class. Both models use TF-IDF ingredient features as input, with product type added as an additional feature for Model 1B.

3.1 Why Multilabel?

Skin type suitability is inherently a multilabel problem because a single product can be suitable for more than one skin type at the same time, a moisturizer may suit both Dry and Sensitive skin, while a serum may work for both Oily and Combination skin. Assigning a single label per product would discard this information and produce an incomplete target. MultiOutputClassifier was used to train one independent binary classifier per skin type label, allowing each label to be predicted independently.

3.2 Why MultiClass?

Primary concern is treated as a single-label multiclass problem because each product in the dataset was assigned exactly one dominant concern based on its afterUse tags. A product cannot simultaneously target Acne as its primary concern and dryness as its primary concern. This makes multiclass classification the appropriate choice.

3.3 Evaluation of Metric Choice

Given the class imbalance present in both tasks, standard accuracy alone is not a reliable evaluation metric. For Model 1A, Hamming Loss is used as the primary metric because it measures how many skin type labels are predicted incorrectly across the five labels, making it suitable for multilabel classification. F1 Weighted and F1 Micro are reported to measure the overall prediction performance, while Per-label F1 is used to identify which skin types are more difficult to predict individually. For Model 1B, Balanced Accuracy is used alongside F1 Weighted and F1 Macro to account for the strong imbalance between dominant classes such as Acne and

minority classes such as Dryness. F1 Macro specifically ensures that minority class performance is not masked by dominant classes.

Metric	Task	Reason
Hamming Loss	1A	Percentage of incorrect individual labels in multilabel setting
F1 Weighted	1A, 1B	Each class contributes proportionally to its frequency
F1 Micro	1A	Treats every label prediction equally
Subset Accuracy	1A	All five labels must be correct to count as correct
Per-label F1	1A	Shows which skin types are hard to predict individually
Accuracy	1B	Standard metric for single prediction per product
Balanced Accuracy	1B	Accounts for class imbalance
F1 Macro	1B	Treats all classes equally regardless of size

3.4 Models Trained

Three classifiers were evaluated for each task, Logistic Regression, LinearSVC and Random Forest . All models were trained with `class_weight='balanced'` to handle class imbalance, and `random_state=42` for reproducibility.

Model 1 : Logistic Regression

We began with logistic regression as a baseline, as it is well known for its effectiveness on sparse high-dimensional text features such as TF-IDF, while also serving as a reference point for comparison with more complex models.

Model 1A skin Type prediction

Label	Dry	Oily	Normal	Combination	Sensitive
F1	0.85	0.95	0.93	0.78	0.86

Hamming-loss	F1 weighted	F1 Micro	Subset Accuracy
0.0973	0.8819	0.8807	0.6451

LR shows great performance and the smallest train-test gap of all three models. It generalizes well but achieves slightly lower overall scores. Combination and Dry are the weakest labels.

Model 1B skin Concern prediction

Class	Acne	Dryness	Eczema	Hyperpigmentation	Redness	Sensitivity
F1	0.93	0.63	0.83	0.81	0.77	0.74

Accuracy	Balanced Accuracy	F1 Weighted	F1 Macro
0.8883	0.9068	0.8915	0.8551

The Logistic Regression model achieved strong overall performance, which means it generalizes well to unseen data. The model performed particularly well in detecting Acne and Eczema, while lower F1-scores for Dryness and Sensitivity indicate that these classes were challenging to classify accurately.

Model 2 : Linear Svc

Linear SVC was selected as it shows strong performance on TF-IDF features, due to its focus on the most informative values while ignoring noise and zero ones

Model 1A skin Type prediction

Label	Dry	Oily	Normal	Combination	Sensitive
F1	0.85	0.95	0.94	0.78	0.86

Hamming-loss	F1 weighted	F1 Micro	Subset Accuracy
0.093	0.8871	0.8859	0.6525

Linear SVC performs similarly to LR shows the most balanced between test and train with a much smaller gap than RF. All five skin types are predicted performing consistently with Oily and Normal performing best and Combination remaining the hardest label across all models.

Model 1B skin Concern prediction

Class	Acne	Dryness	Eczema	Hyperpigmentation	Redness	Sensitivity
F1	0.97	0.67	0.91	0.84	0.80	0.75

Accuracy	Balanced Accuracy	F1 Weighted	F1 Macro
0.8883	0.9068	0.8915	0.8551

The model showed very strong performance indicating good classification and generalization. The model obtained excellent results for Acne and Eczema prediction, while the lower scores for Dryness and Sensitivity suggest that these categories remain more difficult to distinguish.

Model 3 : Random Forest

Random Forest was used as a non-linear model as it captures complex ingredient interactions and non-linear relationships between features and targets which non-linear models can't do.

Model 1A skin Type prediction

Label	Dry	Oily	Normal	Combination	Sensitive
F1	0.88	0.96	0.94	0.79	0.86

Hamming-loss	F1 weighted	F1 Micro	Subset Accuracy
0.0006	0.8995	0.9012	0.7108

RF achieved almost a perfect training performance but dropped on the test, showing a clear sign of overfitting. It learns the training data extremely well but struggles to generalize fully.

Model 1B skin concern prediction

Class	Acne	Dryness	Eczema	Hyperpigmentation	Redness	Sensitivity
F1	0.97	0.70	0.92	0.86	0.81	0.76

Accuracy	Balanced Accuracy	F1 Weighted	F1 Macro
0.8975	0.8166	0.8973	0.8364

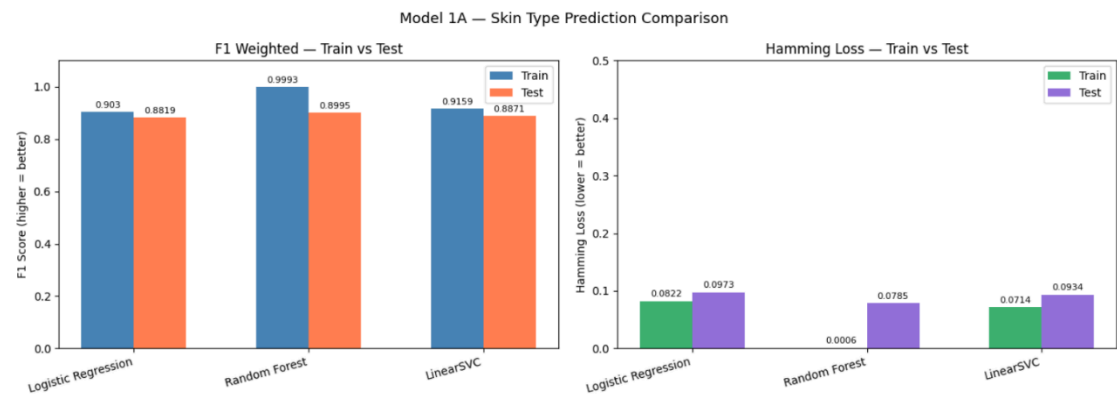
The model achieved the highest overall performance, However, the training results compared to the lower testing performance suggest slight overfitting, although the model still performed very well for Acne and Eczema while Dryness and Sensitivity remained the most challenging classes.

3.5 Model Comparison(Baseline)

Model 1A skin type prediction

Model	Hamming Loss	F1
Logistic Regression	0.0973	0.8819
Linear SVC	0.0934	0.8871
Random Forest	0.0785	0.8995

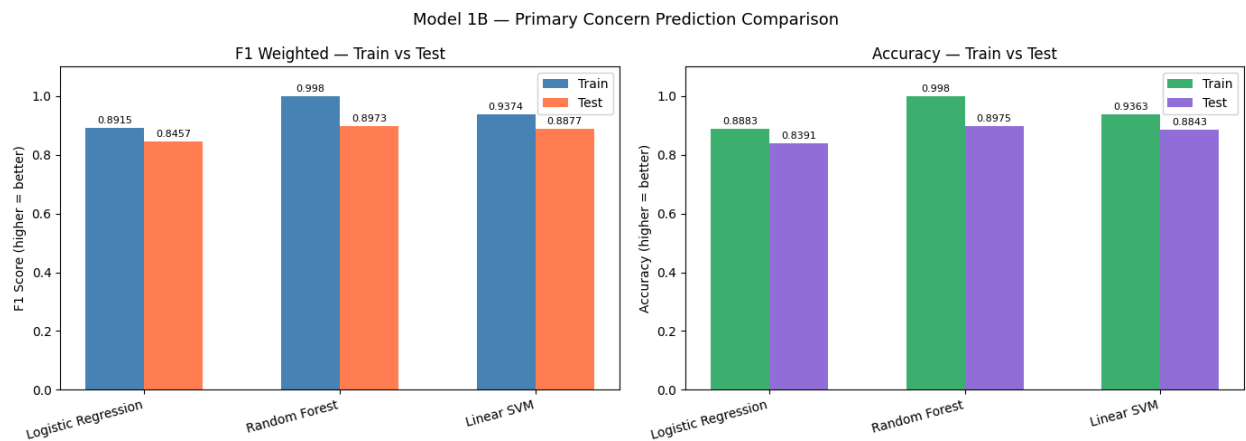
Model	Gap	Overfitting
Logistic Regression	0.0211	no
Random Forest	0.0998	yes
Linear SVC	0.0288	no



All three models perform reasonably well, but they differ in generalization. Random Forest overfits as the gap between training and test performance is too large . Logistic Regression generalizes best with the smallest train-test gap but slightly lower overall performance. Linear SVC shows the best balance, consistent per-label scores, and acceptable overfitting gap, that's why LinearSVC was selected as the final baseline model for skin type prediction. It achieved the best balance between high performance and low overfitting.

Model 1B skin concern prediction

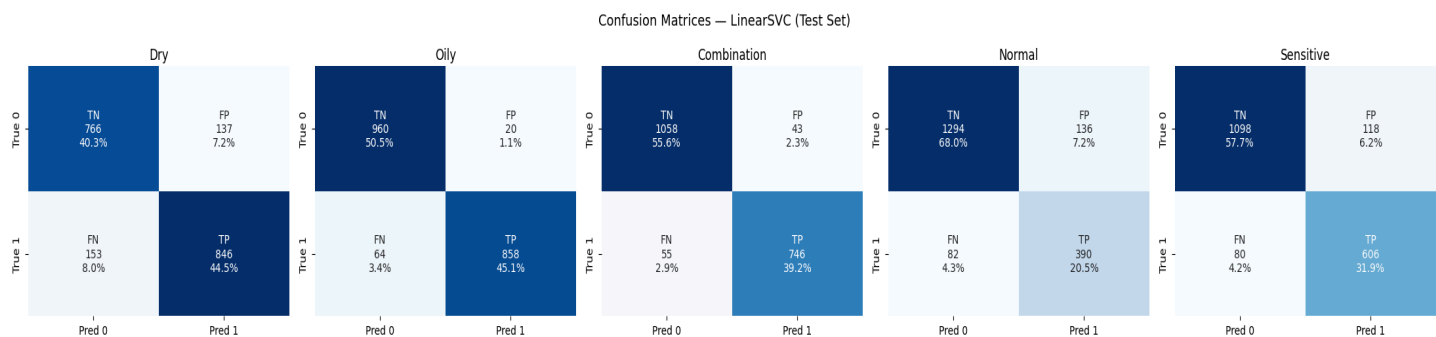
Model	Accuracy	F1	Model	Gap	Overfitting
Logistic Regression	0.89	0.85	Logistic Regression	0.0458	no
Linear SVC	0.88	0.89	Linear SVC	0.0497	no
Random Forest	0.90	0.90	Random Forest	0.1007	yes



Although Random Forest achieved the highest test accuracy and F1-score, it shows signs of overfitting due to the large gap between training and testing performance. Therefore, Linear SVC is selected as the final model because it offers a strong balance between performance and generalization, with stable test results and a smaller overfitting gap compared to Random Forest.

3.6 Confusion Matrix (best baseline)

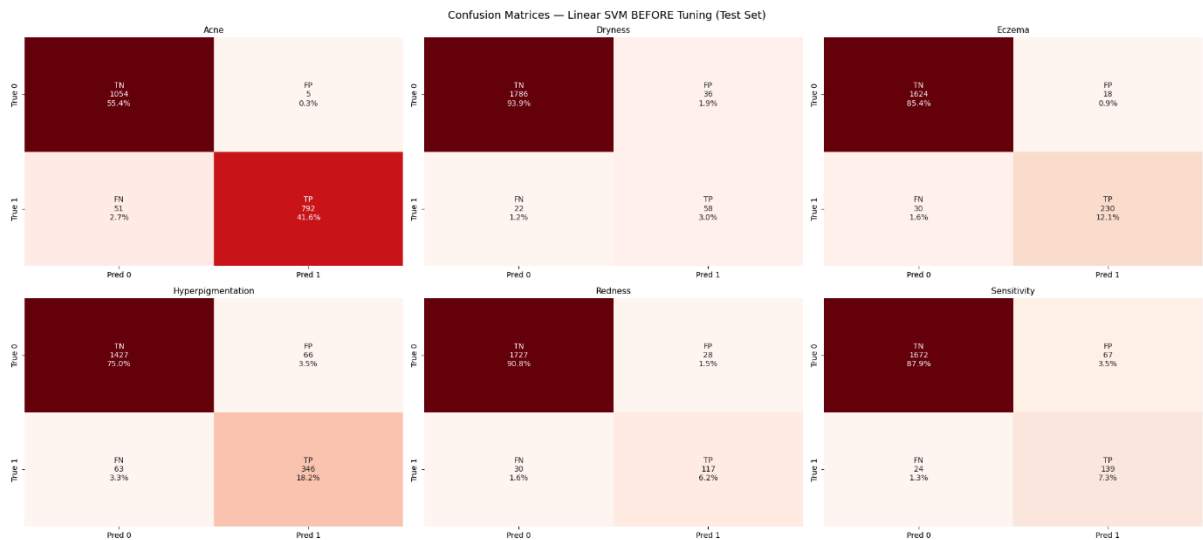
Model 1A skin Type prediction (LinearSVC)



The model is doing well, but performance varies by class. For example, it performs quite strongly on Oily and Combination, where it correctly identifies many true positives and keeps false positives relatively low. The Normal class also shows a high number of true negatives, meaning the model is good at recognizing when a sample is not Normal, but still misses some

false negatives. Classes like Dry and Sensitive show more confusion as the model sometimes mixes them with other skin types.

Model 1B skin concern prediction(Linear SVC)



The Linear SVC model performs well overall, but the results vary between classes. Dryness and Acne show the best performance, with many correct predictions and very few mistakes. Eczema, Redness, and Hyperpigmentation also perform well, although the model still misses or misclassifies some cases. Sensitivity shows slightly more confusion because it has more false positives compared to the other classes. Overall, the model is accurate, but some classes are easier to classify than others.

3.7 Cross Validation

Cross-validation was performed on the top two baseline models . For both models 1A and 1B Linear SVC model shows strong and stable performance during 5-folforcross-validation. The F1-scores remain very consistent across folds, with low standard deviation values, indicating good stability and reliable generalization. The mean F1-scores are high, and the test performance is very close to the cross-validation results, showing minimal overfitting and balanced classification performance on unseen data.

3.8 Hyperparameter Tuning

Model 1A skin Type prediction & Model 1B skin concern prediction

Hyperparameter tuning was performed using RandomizedSearchCV rather than GridSearchCV, because of the computational cost of tuning models on high-dimensional TF-IDF features. A randomized search is faster, finding strong parameter combinations. For Random Forest, the search space included `n_estimators`, `max_depth`, `min_samples_leaf`, `min_samples_split`, and `max_features`. For Logistic Regression and Linear SVC, the regularization parameter `C` was tuned. All searches used 3-fold or 5-fold stratified cross-validation to ensure stable evaluation across folds. The best configuration for each model was selected based on a weighted F1-score on the validation folds.

Model	HyperParameters Model 1A-skin type	HyperParameters Model 1B-skinconcern
Random Forest	<code>n_estimators = 199</code> <code>max_depth = None</code> <code>min_samples_leaf = 8</code> <code>min_samples_split = 4</code> <code>max_features = 0.5</code> <code>class_weight = 'balanced'</code>	<code>n_estimators = 300</code> <code>max_depth = None</code> <code>min_samples_leaf = 2</code> <code>min_samples_split = 4</code> <code>max_features = 'sqrt'</code> <code>class_weight = 'balanced'</code>
Logistics regression	<code>C=4.07</code>	<code>C = 67.32</code>
SVC Linear	<code>C=1.22</code>	<code>C = 2.52</code>

3.9 Model Comparison After Tuning

Model 1A skin type prediction

Model	Hamming Loss	F1
Logistic Regression	0.0928	0.8875
Linear SVC	0.0937	0.8868
Random Forest	0.0737	0.9095

Model	Gap	Overfitting
Logistic Regression	0.0259	no
Linear SVC	0.0296	no
Random Forest	0.0282	no

Before hyperparameter tuning, Random Forest showed severe overfitting with perfect training performance but significantly lower test performance. It was tuned with a reduced search space ($n_iter=10$, $cv=3$) outperforming both Linear SVC and Logistic Regression which used more extensive search ($n_iter=20$, $cv=5$). This suggests RF is the naturally stronger model for this task as the training performance decreased while test performance remained strong. Among all tuned models, Random Forest achieved the best test F1-score, making it the final selected model.

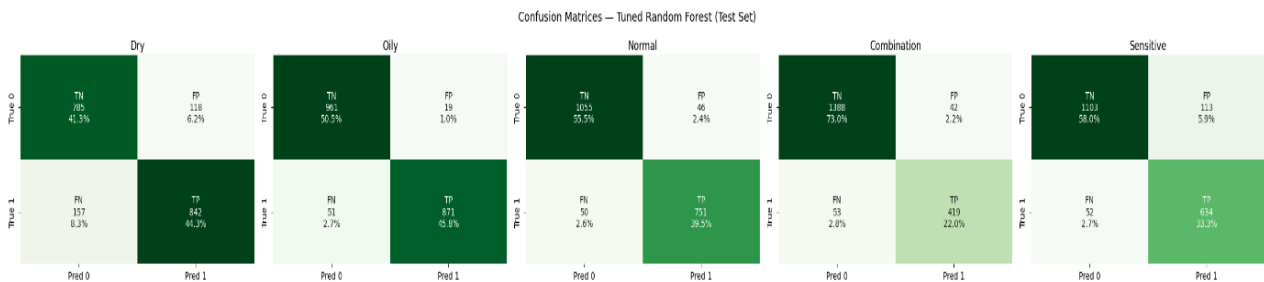
Model 1B skin concern prediction

Model	Accuracy	F1	Model	Gap	Overfitting
Logistic Regression	0.88	0.88	Logistic Regression	0.0892	no
Linear SVC	0.89	0.892	Linear SVC	0.0546	no
Random Forest	0.90	0.90	Random Forest	0.0803	no

Across the three models, Random Forest achieves the highest overall performance with accuracy and F1, followed closely by Linear SVC and Logistic Regression. When looking at the gap between training and test performance, all models show relatively small differences, indicating that none of them are overfitting. Linear SVC being the most stable model, Overall, Random Forest provides the best predictive performance and small gap between test and train, for which it was chosen as the final model.

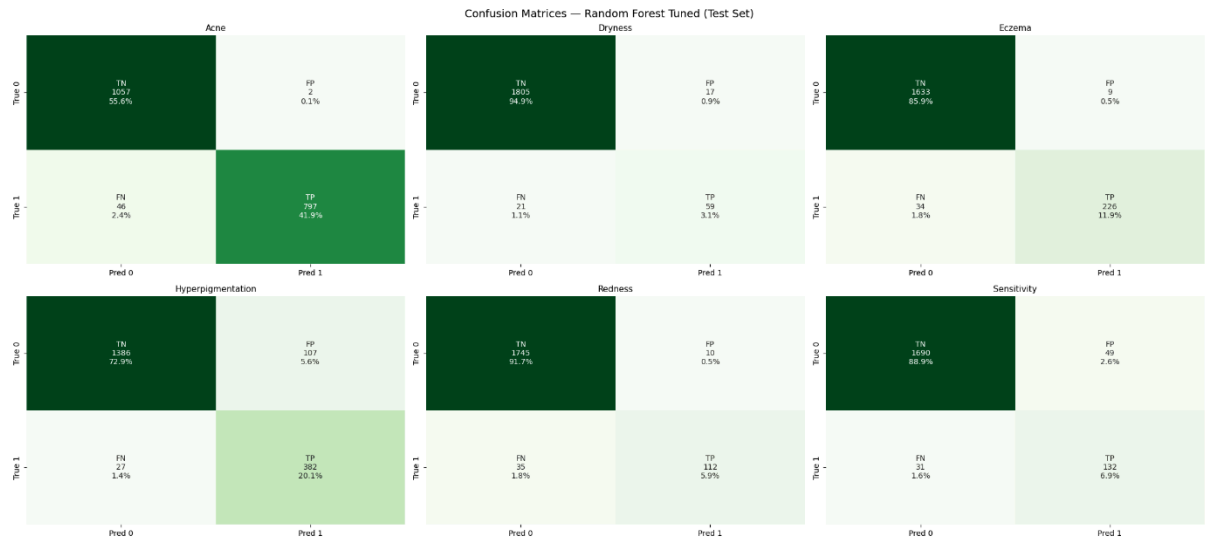
3.10 Confusion matrix (best tuned)

Model 1A skin type prediction (random Forest)



The model demonstrates strong overall performance, particularly excelling in the Normal and Combination categories where true positive rates are notably high. The Dry and Oily classes also show solid diagonal dominance, indicating reliable classification. However, the Sensitive skin type appears to have slightly more misclassifications, suggesting the model struggles somewhat to distinguish it from other categories

Model 1B skin concern prediction(Random Forest)



Random Forest tuned model demonstrates strong overall performance across most skin concern classes. Dryness, Acne, and Redness show particularly high true negative and true positive values, indicating accurate and reliable classification. Eczema and Sensitivity also perform well with relatively few misclassifications. However, Hyperpigmentation shows slightly more false positives compared to the other classes, suggesting the model has more difficulty distinguishing it from similar conditions.

3.11 feature importance

Model 1A skin type prediction

Top 20 most important ingredients (Random Forest):	
Feature	Importance
niacinamide	0.122959
panthenol	0.101211
glycolic_acid	0.099335
retinol	0.065724
retinyl_palmitate	0.059390
salicylic_acid	0.054688
hyaluronic_acid	0.045669
ascorbic_acid	0.032002
sodium_ascorbyl_phosphate	0.022896
bisabolol	0.022231
ascorbyl_tetraisopalmitate	0.021258
ascorbyl_glucoside	0.021244
citric_acid	0.020391
chamomilla_recutita_flower_extract	0.019843
ascorbyl_palmitate	0.015248
sorbitol	0.014824
ethyl_ascorbic_acid	0.013321
tocopherol	0.013101
bakuchiol	0.012666
calendula_officinalis_flower_extract	0.012459

Feature importance analysis was used on the final selected models to identify which ingredients drive predictions most strongly and to see the importance of textual signals. We tried the best model which is Random Forest Tuned. The top 15 features were extracted and used to validate that the model is learning meaningful ingredient-based signals rather than noise, even though some ingredients showed the strongest correlations with the target ,the performance of our model dropped based on F1 score comparison indicating That skin type classification relies on a combination of multiple textual signals of ingredients

The results are shown blow:

Label	Dry	Oily	Normal	Combination	Sensitive
F1	0.75	0.95	0.90	0.874	0.843

Hamming-Io ss	F1 weighted	F1 Micro	Subset Accuracy
0.1217	0.8515	0.8485	0.5778

3.12 final choice

The results confirm that ingredient lists carry meaningful signals for predicting skin concern and skin type, especially for dominant classes. All three classifiers showed strong performance, good generalization, and stability, particularly after tuning. Cross-validation further confirmed the stability of the models across folds.

As a result, Random Forest was chosen as the final model, as it was able to capture the complexity and non-linearity of the data very well.

The tables below shows the final result:

Model 1A skin type prediction

Label	Dry	Oily	Normal	Combination	Sensitive
F1	0.86	0.96	0.94	0.90	0.89

Hamming-loss	F1 weighted	F1 Micro	Subset Accuracy
0.00737	0.9095	0.9094	0.7287

Model 1B skin concern prediction

Class	Acne	Dryness	Eczema	Hyperpigmentation	Redness	Sensitivity
F1	0.97	0.76	0.91	0.86	0.83	0.76

Accuracy	Balanced Accuracy	F1 Weighted	F1 Macro
0.8980	0.8430	0.9000	0.8485

Conclusion

This project set out to answer a simple but important question: can we help users evaluate the safety of their skincare before skin damage occurs? The answer, based on our work, is yes.

We developed a two-part AI-based Skincare Routine Risk Assessment System. The first part detects ingredient compatibility conflicts using a tuned Random Forest classifier that achieved a recall of 0.99, an F1-score of 0.9832, and a ROC-AUC of 0.9947, with no signs of overfitting confirmed by 5-fold cross-validation. The second part predicts skin type suitability and primary skin concern from ingredient lists, also using a tuned Random Forest, achieving an F1-weighted score of 0.91 for skin type and 0.90 for skin concern prediction.

The CRISP-DM methodology guided every phase of the project, from business understanding and data collection through exploratory analysis, preprocessing, modeling, and evaluation. Key technical decisions including SMOTE for class imbalance, feature engineering of 11 interaction variables, TF-IDF vectorization of ingredient text, and careful handling of missing values were all grounded in both data findings and dermatological knowledge.

The system does not replace dermatologists. It serves as a preventive tool that educates users, flags risky combinations, and helps people make more informed choices about the products they apply to their skin every day.